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Where is the gas? Exploring non-detections of CO(1–0) emission in high-redshift galaxies using the Schmidt-Kennicutt Relation

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ABSTRACT

Galaxy assembly in the early universe can be probed using the Schmidt-Kennicutt (SK) relation, where the physical mechanisms that drive star formation can be constrained by determining the SK relation's slope and normalization. We attempt to measure the SK relation for 5 gravitationally lensed galaxies at "Cosmic Noon" ($z \sim 2$–3) using a hybrid spatially resolved star formation rate (SFR) tracer constructed with data from the *Hubble Space Telescope (HST)* and Atacama Large Millimeter/Submillimeter Array (ALMA), in conjunction with Karl G. Jansky Very Large Array (VLA) observations of the CO(1–0) line. We fit existing multiband observations to a spectral energy distribution (SED) using Prospector to derive key physical parameters. These parameters include the SFR and stellar mass, which are used to scale the hybrid SFR tracer as well as a metallicity and "Main Sequence" (MS) offset dependent CO–to–H_2 conversion factor. Since we obtain mostly non-detections in the CO(1–0) observations, we use the hybrid resolved SFR map and various assumed forms of the SK relations (varying slope and normalization) to construct CO(1–0) line emission models and simulated VLA observations to explain the non-detections. While some observations were of insufficient depth due to poor constraints on the spatial extents of the emission, two galaxies appear to have higher star formation efficiencies (SFEs) than typically assumed for "normal"/MS galaxies, as well as higher values of the CO–to–H_2 conversion factor than typically assumed for starbursts with those SFEs, making their CO(1–0) emission particularly dim. These results illustrate the diverse factors that complicate resolved CO mapping experiments at Cosmic Noon and may cause many of the CO non-detections seen in telescope archives.

Keywords: High-redshift galaxies (734); Molecular gas (1073); Star formation (1569)

1. INTRODUCTION

Studies of high redshift galaxies are essential in order to constrain possible mechanisms of galaxy formation and evolution. In terms of overall star formation history, "Cosmic Noon" at z $\sim 2-3$ (referred to as high-redshift hereafter) is the epoch at which the star formation rate density (SFRD) was at its peak, (e.g., S. J. Lilly et al. 1996, P. Madau & J. M. Shull 1996, P. Madau et al. 1998, or N. M. Förster Schreiber & S. Wuyts 2020). Most galaxies assembled large fractions of their stellar mass during this time period (e.g., R. S. Somerville & R. Davé 2015), and therefore understanding this period's high SFR is critical to understanding the population of galaxies that exists today.

An overarching framework for understanding the interplay of star formation and galaxies' gas content is the Schmidt-Kennicutt (hereafter SK) relation, first theorized by M. Schmidt (1959) and observed by R. C. Kennicutt (1998a). The SK relation connects the surface density of the gas mass (Σ_{gas}) with the surface density of the star formation rate (SFR; Σ_{SFR}) as a power law, which can be expressed as:

$$\log(\Sigma_{SFR}) = A + n \times \log(\Sigma_{gas}). \quad (1)$$

The SK relation's slope, n, is tied to the physical mechanisms that link star formation with their birth clouds such as the timescales set by self-gravity ($n = 1.5$; e.g., B. G. Elmegreen 2002), turbulence in self-regulated disks ($n \sim 1.5$–2; e.g., E. C. Ostriker & R. Shetty 2011; M. R. Krumholz & C. F. McKee 2005), cloud-cloud collisions ($n \gtrsim 3$; e.g., M. Zamora-Avilés et al. 2012; S. Horie et al. 2024), and the result of multiple processes acting at different timescales in self-regulated star formation ($n = 1$; e.g., V. A. Semenov et al. 2019). The SK relation's offset constant, A, is frequently considered a probe of the underlying star formation efficiency (SFE; where SFE = SFR/M_{gas}), albeit with some degeneracies based on the relation's slope (e.g., A. K. Leroy et al. 2008).

Despite the SK relation's importance for understanding star formation processes, relatively few measurements of the SK relation exist for high-redshift galaxies. The chief observational challenge for high-redshift measurements of the SK relation is spatially resolved detections of the gas and SFR tracers. While luminosity-luminosity correlation versions of the SK relation are more common at high redshift (e.g., E. Daddi et al. 2010; R. J. Ivison et al. 2011; M. T. Sargent et al. 2014; M. Dessauges-Zavadsky et al. 2015; A. Kirkpatrick et al. 2019), studies of low-redshift galaxies find different forms of the SK relation between the luminosity-luminosity correlations $n \sim 1$– 1.5; (e.g., P. M. Solomon & L. J. Sage 1988; Y. Gao & P. M. Solomon 2004), disk-wide average surface brightnesses $n \sim 1.4$; (e.g., R. C. Kennicutt 1998b; M. Murgia et al. 2002; A. Leroy et al. 2005; L. H. Wei et al. 2010), and spatially resolved surface brightness measurements $n \sim 1$ with individual galaxies differing slightly; (e.g., A. K. Leroy et al. 2013; T. Wong & L. Blitz 2002; S. Boissier et al. 2003; K. F. Schuster et al. 2007; R. C. Kennicutt et al. 2007; F. Bigiel et al. 2008). So far, only a handful of galaxies at $z \gtrsim 1.5$ have sufficiently resolved observations to measure pixel-to-pixel variations in the SK relation within individual galaxies: C. E. Sharon et al. (2013), R. Genzel et al. (2013), T. D. Rawle et al. (2014), J. A. Hodge et al. (2015), R. Cañameras et al. (2017), K. Tadaki et al. (2018), C. E. Sharon et al. (2019), T. Tsukui et al. (2023), Y.-C. Chen et al. (2024), A. Zanella et al. (2024), L. Vallini et al. (2024), S. Arriagada-Neira et al. (2025), J. S. Gómez et al. (2025), D. Vizgan et al. (2026).

With the relatively small sample of high-redshift sources with adequately resolved observations, source selection effects and methodological choices significantly affect the measured SK relation. High-redshift star-forming galaxies have historically been identified in the UV (often characterized as "normal" MS disk galaxies) and submillimeter/far-infrared regimes (often characterized as dusty starbursts analogous to local ultra-luminous/luminous IR galaxies; U/LIRGs). While recent work suggests that there is significant overlap between the populations of dusty star-forming galaxies (DSFGs) and UV-bright galaxies (at least among more massive galaxies; e.g., R. R. Lindner et al. 2012; R. Decarli et al. 2014; F. Walter et al. 2014; S. Wuyts et al. 2011), their method of selection often results in complementary tracer selection for measuring the SFR axis of the SK relation. Even within a single galaxy, differing methods of extinction correction have been shown to produce different slopes for the SK relation (R. Genzel et al. 2013), which means incomplete characterization of both the obscured and unobscured star formation as a function of galaxy population potentially introduces systematic effects in the measurement of the high-redshift SK relation. In addition, systematic effects can also be introduced into the molecular gas axis of the SK relation. The excitation of the observed carbon monoxide (CO) rotational line can produce flatter slopes for higher-J CO lines (e.g., D. Narayanan et al. 2008; M. R. Krumholz & T. A. Thompson 2007; D. Narayanan et al. 2011; T. R. Greve et al. 2014). The choice in CO-to-H_2 conversion factor, α_{CO}, often differs between galaxies populations, with $\alpha_{CO} = 0.8\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ used for starbursts (e.g., N. Z. Scoville et al. 1991; D. Downes et al. 1993; P. M. Solomon et al. 1997; D. Downes & P. M. Solomon 2003; D. S. Meier et al. 2010), and $\alpha_{CO} = 4.36\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ for the Milky Way galactic disks and other local disky galaxies (e.g., R. B. Larson 1981; A. K. Leroy et al. 2011; A. W. Strong & J. R. Mattox 1996). However, the CO-to-H_2 conversion factor is also known to vary with metallicity and temperature (e.g., C. D. Wilson 1995; N. Arimoto et al. 1996; A. Leroy et al. 2006; A. K. Leroy et al. 2011; R. Genzel et al. 2012; D. Narayanan et al. 2011; B. Magnelli et al. 2012) and varies with physical conditions within a single galaxy (K. M. Sandstrom et al. 2013; I.-D. Chiang et al. 2024).

In order to better characterize the high-z resolved SK relation, we observe a sample of gravitationally lensed galaxies at $z \sim 2$–3 that have a combination of existing photometry in the continuum far-IR and UV, and/or existing constraints on the molecular gas. We target strongly lensed galaxies in order to more easily spatially resolve the high-redshift targets. These galaxies' existing data enabled observations of both obscured and unobscured SFR tracers, attempted detections of the CO(1–0) line, and more informed choices in α_{CO} based on galaxy characteristics like metallicity; in this way we are able to avoid many of the biasing factors that may affect analysis of the high-redshift SK relation. In Section 2, we describe our sample of 5 galaxies. In Section 3 we describe data obtained from the VLA, ALMA, and *HST*, as well as other archival data using in spectral energy distribution (SED) fitting. In Section 4, we describe the results of our observations as well as any unique details about their observations and data processing. In Section 5, we describe the photometric fitting results used (primarily) to construct the SFR maps. Since the CO observations yielded weak or non-detections, we analyze how our limits can constrain some of the galaxies' properties by comparing them to the SK relation in 6 and comparing them to simulations of the CO emission in Section 7.

For the remainder of the work, we assume a flat cosmology with $H_0 = 69.6$ km/s/Mpc and $\Omega_{vac} = 0.714$. We also assume a Kroupa IMF (P. Kroupa 2001).

2. SOURCES

The sample comprises 5 gravitationally lensed galaxies at z $\sim$ 2–3 (Table 1). Most of them have existing integrated CO(1–0) (ν_{rest} = 115.2712 GHz) measurements from lower-resolution configurations of the VLA or constraints on higher-excitation CO lines, like CO(2–1) or CO(3–2). Two galaxies, the Cosmic Eye and cB58, were chosen for high-spatial resolution CO mapping due to previous CO(1–0) detections in D. A. Riechers et al. (2010). The Cosmic Horseshoe, 8:00arc, and A68-h7 had previous low-resolution CO(3–2) detections as shown in A. Saintonge et al. (2013) for the first two and M. Dessauges-Zavadsky et al. (2015) for A68-h7. These galaxies are all FIR-detected MS galaxies (A. Saintonge et al. 2013; M. Dessauges-Zavadsky et al. 2015) where magnification by gravitational lensing affords a good opportunity to study resolved star formation and molecular gas around cosmic noon. A summary of the available low-resolution CO detections is given in Table 2.

3. DATA REDUCTION AND PROCESSING

3.1. *Karl G. Jansky Very Large Array (VLA)*

We acquired VLA observations of the CO(1–0) lines for these galaxies in projects 14B-278, 16A-077, and 18A-114 respectively (PI: C. Sharon). Data was primarily acquired in C configuration in 14B-278 and 16A-077, and primarily in D configuration for 18A-114. The first five data tracks in 14B-278 were acquired in DnC configuration and the last six data tracks in 16A-077 were acquired in CnB configuration. A summary of the observation dates, integration times, calibrators, and correlator setup is presented in Table 3. Additional details for specific objects are described in Section 4.

For our observations, the flux calibrator was chosen from the standard list of VLA calibrators and model fluxes are taken from the CASA package's default 'Perley-Butler' standard. We also performed bandpass calibration using the flux calibrator. Phase and amplitude fluctuations were tracked using a nearby quasar. The flux and phase calibrator for each source is listed in Table 3. We observed with the WIDAR correlator using the widest subbands (128 MHz) and the lowest spectral resolution (1 MHz channels), which uses dual polarization. We observed two sidebands: one tuned such the redshifted CO(1–0) line frequency is centered in a central subband, and the second tuned to sample continuum as near as possible to the first subband given correlator tuning restrictions.

The VLA data was calibrated using a VLA pipeline on the Common Astronomy Software Applications (CASA) version 5.6.2. Data showing rapid variations in amplitude and phase measurements for the calibrators were further flagged to improve signal detection and remove imaging artifacts. Data taken in poor weather conditions (i.e. with high variance in phase measurements) were particularly scrutinized, and antennas showing large delays or high phase variation were flagged.

The calibrated VLA data was then deconvolved with `tclean` in CASA, using the Hogbom algorithm (J. A. Högbom 1974) and natural weighting. We `CLEAN` on the source by restricting model components to source arcs visible in the *HST* and/or ALMA data, as well as any bright foreground continuum sources. We iteratively `CLEAN` data cubes of different spectral resolutions, starting with 100 km s^{-1} channels with one channel centered at the expected observed frequency of the CO(1–0) line (ν_{obs}). We then dither the center frequency slightly about ν_{obs} and increase the channel width to maximize the detection SNR and to confirm non-detections whenever applicable.

3.2. *Atacama Large Millimeter Array (ALMA)*

ALMA data used in this project were obtained as part of program 2019.1.01148.S (PI: C. Sharon) and from various archival programs (listed in Table 4). We downloaded the raw data tracks along with observatory generated flags and calibration scripts from the ALMA archive[2], running them on the suggested CASA versions. When imaging, we used CASA version 5.6.2. The data required no additional manual flagging and were imaged similarly to the VLA data, except we used multi-frequency synthesis (`mfs`) spectral gridding in CASA. Additional details for specific objects are described in Section 2.

[2] https://almascience.nrao.edu/aq/

Table 1. Lensed galaxy sample

Source Name	α	δ	z_{spec}	z_{spec} Ref.	Magnification	Mag. Ref.	Z	Z Ref.
Cosmic Eye	$21^h35^m12.7^s$	$-01^d01^m42.9^s$	3.074	a	30.0 ± 3.3	f	8.64 ± 0.14	h
8:00arc	$00^h22^m40.9^s$	$+14^d31^m14.0^s$	2.730	b	$12.3^{+15}_{-3.6}$	b	8.52 ± 0.10	h
cB58	$15^h14^m22.2^s$	$+36^d36^m25.2^s$	2.729	c	31.8 ± 8.0	g	8.36 ± 0.14	c
Cosmic Horseshoe	$11^h48^m33.1^s$	$+19^d30^m03.2^s$	2.379	d	30.0 ± 5.0	d	8.36 ± 0.13	i
A68-h7	$00^h37^m01.4^s$	$+09^d10^m22.3^s$	2.15	e	3	e	8.52^*	

References—[a] I. Smail et al. (2007) [b] S. S. Allam et al. (2007) [c] H. I. Teplitz et al. (2000) [d] V. Belokurov et al. (2007) [e] M. Dessauges-Zavadsky et al. (2015) [f] S. Dye et al. (2007) [g] S. Seitz et al. (1998) [h] J. Richard et al. (2011) [i] K. N. Hainline et al. (2009)

* Assumed metallicity using average population relations in T.-T. Yuan et al. (2013); see Section 6.2.

Table 2. Existing low-J CO detections in the literature

Source Name	CO line	$S\Delta\nu$ (Jy km/s)	Reference
Cosmic Eye	CO(1–0)	0.077 ± 0.013	D. A. Riechers et al. (2010)
	CO(3–2)	0.50 ± 0.07	K. E. K. Coppin et al. (2007)
8:00arc	CO(3–2)	0.85 ± 0.24	A. Saintonge et al. (2013)
cB58	CO(1–0)	0.052 ± 0.013	D. A. Riechers et al. (2010)
	CO(3–2)	0.37 ± 0.08	A. J. Baker et al. (2004)
Cosmic Horseshoe	CO(3–2)	0.44 ± 0.18	A. Saintonge et al. (2013)
A68-h7	CO(3–2)	1.12 ± 0.22	M. Dessauges-Zavadsky et al. (2015)

Table 3. VLA Observations

Source Name	Phase Cal.	Flux Cal.	Date	$t_{\rm int}$ (hr)	Time of Day	Weather Conditions	Phase Cal. S_ν (Jy)	ν^a (GHz)
Cosmic Eye	J2134-0153	3C48	Sep 19 2014	1.19	Night	Mostly Clear	2.39	32.28
			Sep 26 2014	0[b]				
			Sep 29 2014	1.95	Night	Clear	2.53	
			Oct 02 2014	1.95	Night	Clear	2.49	
			Oct 03 2014	1.95	Night	Clear	2.47	
			Oct 03 2014	1.19	Night	Clear	2.65	
			Oct 10 2014	1.00	Night	Cloudy	2.66	
			Oct 16 2014	1.00	Night	Mostly Clear	2.58	
			Oct 17 2014	0[b]				
			Oct 20 2014	0.24	Night	Clear	2.57	
			Oct 24 2014	0.24	Night	Mostly Clear	2.58	
			Oct 26 2014	0.66	Night	Mostly Clear	2.55	
			Oct 28 2014	0.85	Night	Mostly Clear	2.64	
			Nov 08 2014	1.95	Sunset - Night	Clear	2.44	32.15
			Nov 08 2014	0.24	Night	Clear	2.52	32.28
			Nov 13 2014	0.44	Night	Clear	2.38	
			Dec 02 2014	0.44	Evening	Overcast	2.39	
			Dec 28 2014	0.66	Evening	Mostly Clear	2.16	
			Jan 04 2015	1.19	Afternoon	Clear	2.10	
			Feb 12 2016	0.85	Afternoon	Clear	1.40	
			Feb 21 2016	0.44	Afternoon	Mostly Clear	1.49	
			Feb 25 2016	0[c]				
			Mar 18 2016	1.19	Morning	Clear	1.66	
			Mar 19 2016	0.66	Morning	Clear	1.60	
			Mar 27 2016	1.95	Morning	Clear	1.64	
			Apr 07 2016	1.00	Morning	Overcast	1.63	
			Apr 14 2016	1.00	Sunrise	Clear	1.68	
			Apr 19 2016	1.95	Sunrise	Clear	1.66	
			Apr 21 2016	0[c]				
			Apr 25 2016	0[b]				
			Apr 28 2016	1.95	Morning	Clear	1.60	
			Apr 28 2016	0[b]				
			Apr 29 2016	1.95	Morning	Mostly Clear	1.59	
			May 02 2016	0[b]				
			Total:	*30*				
8:00arc	J0022+0608	3C48	Oct 10 2018	1.06	Morning	Clear	0.17	31.66
			Oct 21 2018	1.04	Night	Mostly clear	0.17	31.18
			Oct 23 2018	1.04	Night	Partly Cloudy	0.16	31.66
			Oct 23 2018	1.06	Night	Cloudy	0.24	
			Oct 30 2018	1.06	Night	Partly Cloudy	0.17	
			Total:	*5.26*				
cB58	J1506+3730	3C286	Mar 1 2016	0.49	Sunrise	Partly Cloudy	1.45	32.53
			Apr 1 2016	1.64	Sunrise	Cloudy + Some Snow	1.44	
			Apr 10 2016	1.23	Sunrise	Partly Cloudy	1.47	
			Apr 14 2016	1.99	Sunrise	Clear	1.43	
			Apr 16 2016	1.99	Sunrise	Mostly Clear	1.43	
			Apr 17 2016	1.99	Sunrise	Mostly Clear	1.38	
			Apr 20 2016	1.99	Morning - Sunrise	Clear	1.39	
			Apr 26 2016	1.99	Early Morning	Clear	1.34	
			May 14 2016	0.91	Early Morning	Mostly Clear	1.32	
			Total:	*14.1*				
A68-h7	J0022+0608	3C48	Oct 09 2018	1.31	Mostly Clear	Night	0.16	37.2

[a] Empty values in this column indicate that the value is the same as the preceding row

[b] Data missing due to missing flux calibration scan.

[c] Data missing due to incomplete/noisy scan.

Table 4. ALMA Observations

Name	PI	Project Code	Scheduling Block[a]	Flux Cal.	Bandpass Cal.	Phase Cal.	PWV (mm)	Phase Cal. Flux (Jy)	Phase Cal. Freq. (GHz)	$t_{\rm int}$ (hours)
Cosmic Eye	K. Coppin	2019.1.01642.S	X17d0	J2253+1608	J2253+1608	J2134-0153	3.18	1.13	95.1	0.53
	A. Man	2019.1.00790.S	X470a	J2253+1608	J2253+1608	J2134-0153	1.8	1.5	87.3	0.7
		2019.1.00790.S	X4b1c	J2253+1608	J2253+1608	J2134-0153	1.7	1.58	87.3	0.7
		2019.1.00790.S	X8a43	J2253+1608	J2253+1608	J2134-0153	1.4	1.56	87.3	0.7
	K. Coppin	2012.1.00008.S	X11c0	Titan	J2258-2788	J2148+0657	5.65	2.55	91.9	0.77
		2012.1.00008.S	X18f5	Ceres	J2148+0657	J2148+0657	6.73	3	91.9	0.77
		2012.1.00008.S	X1491	Ceres	J2148+0657	J2148+0657	7.9	3.62	91.9	0.68
		2012.1.00008.S	X16c	Ceres	J2148+0657	J2148+0657	2.1	3.14	91.9	0.77
		2012.1.00008.S	Xfa0	Ceres	J2148+0657	J2134-0153	1.46	1.41	91.9	0.77
	N. Tohru	2013.1.00139.S	**X91cd**	Neptune	J2148+0657	J2148+0657	1.038[b]	0.81	353.1	0.5
		2013.1.00139.S	**X22f7**	Ceres	J2148+0657	J2148+0657	1.038[b]	0.81	353.1	0.85
		2013.1.00139.S	X2434	Ceres	J2148+0657	J2148+0657	0.236	0.63	461.1	0.03
8:00arc	S. Malhotra	2013.1.00262.S	**X4557**	Uranus	J2258-2788	J0019+2021	2.19	0.31	231.4	0.16
		2015.1.01157.S	**X2c5**	J0006-0623	J0006-0623	J0056+1625	3.1	0.1	231.4	0.74
		2015.1.01157.S	**X5b93**	Pallas	J0006-0623	J0010+1058	1.2	0.41	231.4	0.74
		2015.1.01157.S	Xb07c	Pallas	J0006-0623	J0010+1058	2.31	0.87	133	0.79
	G. Drouart	2016.1.01231.S	X1176	J0006-0623	J0006-0623	J0010+1724	1.93	0.52	99.4	0.45
	A. Man	2019.1.00790.S	Xc1b6	J0006-0623	J0006-0623	J0010+1724	1.21	0.6	95.3	0.49
cB58	S. Malhotra	2013.1.00262.S	**X15a4**	Titan	J1550+0527	J1419+2706	0.8	0.22	224.6	0.18
		2015.1.01157.S	X18be	Titan	J1550+0527	J1453+2648	1.1	0.21	138.6	0.48
		2015.1.01157.S	**X5468**	Titan	J1550+0527	J1453+2648	1.1	0.15	224.4	0.59
	C. Sharon	2019.1.01148.S	X7310	J1337-1257	J1337-1257	J1522+3144	0.72	0.05	336.5	0.25
	A. Man	2019.1.00790.S	X15076	J1550+0527	J1550+0527	J1522+3144	3.9	0.12	95.4	0.73
		2019.1.00790.S	X8505	J1550+0527	J1550+0527	J1522+3144	1.5	0.11	95.4	0.73
		2019.1.00790.S	Xc9c2	J1550+0527	J1550+0527	J1522+3144	3.5	0.13	95.4	0.73
		2019.1.00790.S	Xa523	J1550+0527	J1550+0527	J1522+3144	3.8	0.11	95.4	0.73
		2019.1.00790.S	X12d6a	J1550+0527	J1550+0527	J1522+3144	3.4	0.12	95.4	0.73
		2019.1.00790.S	X2678	J1550+0527	J1550+0527	J1522+3144	3.6	0.11	95.4	0.73
		2019.1.00790.S	X2a03	J1550+0527	J1550+0527	J1522+3144	3.6	0.12	95.4	0.73
Cosmic Horseshoe	C. Sharon	2019.1.01148.S	**X485e**	J1058+0133	J1058+0133	J1157+1638	0.18	0.06	350.5	0.6
		2019.1.01148.S	**Xe042**	J1058+0133	J1058+0133	J1148+1840	0.41	0.09	350.5	0.6
	G. Drouart	2016.1.01231.S	X3ca2	J1058+0133	J1058+0133	J1148+1840	0.74	0.24	90.3	0.4
	S. Malhotra	2013.1.00262.S	X611d	Callisto	J1058+0133	J1215+1654	2.41	0.21	239.9	0.17

[a] The bolded scheduling blocks are those data there were used for spatially resolved continuum IR maps to construct SFR maps in Section 6.1.

[b] These PWV were averaged across the two scheduling blocks.

3.3. *Hubble Space Telescope (HST)*

Archival data was downloaded from the *Hubble* Legacy Archive[3] and reprocessed for this analysis. We used stage 3 data wherever available, which has been pre-processed by the pipeline and includes corrections for geometric distortion, combination of dithered observations, flux calibration, and cosmic ray corrections. We used data from the Advanced Camera for Surveys (ACS), Near Infrared Camera and Multi-Object Spectrometer (NICMOS), Wide Field Planetary Camera 2 (WFPC2), and Wide Field Camera 3 (WFC3) whenever available for our sources. The specific projects, PI names, and filters used are listed in Table 5.

We obtained photometry for multiple bands by creating apertures through visual inspection. We perform a local background subtraction from near the sources' apertures to obtain new fluxes (listed in Table 5).

In order to create composite SFR maps (Section 6.1), we regridded and smoothed the *HST* maps to match the pixel sizes and resolution of the ALMA data using `imsmooth` in CASA. The FWHM of the *HST* point-spread function (PSF) is on the order of $\sim 0.1''$, which is significantly smaller than the corresponding ALMA observations' synthesized beam FWHMs. Therefore precise modeling of *HST* PSF is not required. Using the *HST* handbook relevant to the appropriate instrument (WFC3: M. H. Wong et al. 2010; ACS: A. Suchkov 2001; WFPC2: M. McMaster & et al. 2008) we extracted PSF FWHMs for filters from the different instruments and observed wavelengths as input to `imsmooth`.

3.4. *Additional archival data*

Whenever possible, we have used literature reported optical and far-infrared data to constrain spectral fits. We identified several additional bands from the Sloan Digital Sky Survey (SDSS) and *Spitzer* without published photometry, so we describe those observations and data processing steps here.

Archival SDSS data was downloaded from the SDSS SkyServer[4]. All our objects can be identified in images from the SDSS Legacy Survey with the exception of A68-h7, which was a non-detection in the SDSS-III survey.

Archival *Spitzer* data was downloaded from the Spitzer Heritage Archive[5]. We downloaded data processed at level 2 ("Post-Basic Calibrated Data"), which are data that are stacked individual exposures after being calibrated using the *Spitzer* data calibration pipeline.

Since we only use these datasets for photometry, we perform a local background subtraction from near the sources' apertures (rather than flat-fielding the whole image) to obtain fluxes that are accurate to $\sim 10\%$ when also considering flux calibration, aperture choice, and statistical uncertainties[6]. Due to seeing limits on ground-based facilities, some of our sources have flux contamination in the SDSS data due to nearby objects such as the foreground lensing galaxies, although the level of contamination varies by source. Similarly, the diffraction-limited Spitzer images also have flux contamination from the foreground lens for some of our strongly lensed sources. The most affected source is the Cosmic Eye. However, these data are still included in the spectral fit as our scientifc objects dominate the flux at these bands (our strongly lensed sources are roughly 2-3 times brighter than their lens at the longest wavelength *HST* bands) and the additional flux contamination at these bands likely do not affect the spectral fits much more than their existing uncertainties. Specific concerns regarding flux contamination for individual sources is described in Section 4.

4. OBSERVATION RESULTS FOR INDIVIDUAL SOURCES

Our new CO(1–0) observations primarily resulted in non-detections for these galaxies (Fig. 1), both for the higher resolution observations of previously CO(1–0)-detected sources and for the new low-resolution observations of sources that only had higher-J CO detections previously. The only exception is the Cosmic Eye, where we obtained a marginal detection. While not observed at the VLA, we also include the Cosmic Horseshoe in this work due to the availability of spatially resolved *HST* and ALMA data making it valuable for predictive exercises (see Section 7).

4.1. *Cosmic Eye*

The Cosmic Eye has existing unresolved CO(1–0) line emission from the VLA (D. A. Riechers et al. 2010) and CO(3–2) line emission from the six-element IRAM Plateau de Bure Interferometer (K. E. K. Coppin et al. 2007). We use VLA data from two projects, 16A-077 and 14B-278 (PI: C. Sharon), to better resolve the CO(1–0) emission. J2134-0153 was used as the amplitude and phase calibrator, and 3C48 was used as the flux and bandpass calibrator. Most of the data is in C-configuration, with the visibilities associated with longer baselines in CnB clipped before imaging. Data taken on

[3] https://hla.stsci.edu/
[4] https://skyserver.sdss.org/
[5] https://irsa.ipac.caltech.edu/applications/Spitzer/SHA/
[6] https://catcopy.ipac.caltech.edu/dois/doi.php?id=10.26131/IRSA486

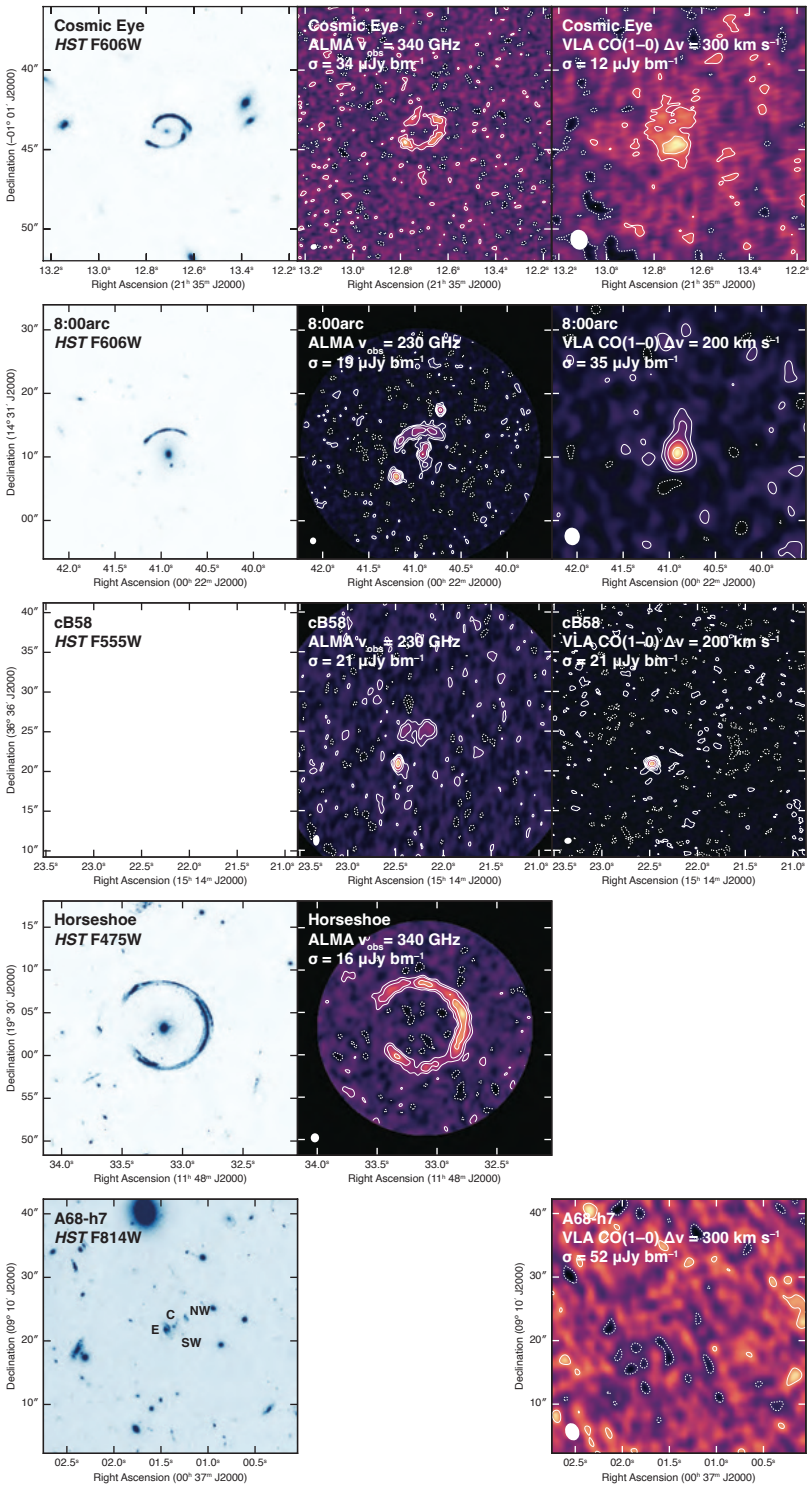


Figure 1. *HST* (left column), ALMA continuum (center column), and VLA CO(1–0) images (right column) for the five lensed galaxies in this study. We show images from the *HST* bands and ALMA datasets that are used in the construction of the SFR maps in Section 6.1; the *HST* bands and ALMA observing frequencies are listed in each panel. For the ALMA and VLA observations, the synthesized beam FWHM is shown as a white ellipse in the lower left corners of each panel, and the image noise is also listed. For the ALMA maps, the contours are *powers* of $\pm 2 \times \sigma$ (i.e. $\pm 2\sigma$, $\pm 4\sigma$, $\pm 8\sigma$, etc.), while for the VLA maps, the contours are *multiples* of $\pm 2\sigma$ (i.e. $\pm 2\sigma$, $\pm 4\sigma$, $\pm 6\sigma$, etc.); negative contours are dotted. In addition, for the VLA data, we list the velocity width integrated over to produced the CO(1–0) maps. Note that for the 8:00arc, the bulk of the VLA emission is continuum emission produced by the central lens. In the *HST* image for A68-h7 we also label the four components (eastern, central, northwestern, and southwestern) that may be the source of the unresolved infrared continuum emission (either as singular objects, or in aggregate as a potential ongoing merger; P. Sklias et al. 2014; M. Dessauges-Zavadsky et al. 2015).

Table 5. *HST* data and fluxes used

Source Name	Filter	λ_{eff}[a] (nm)	$\Delta\lambda_{eff}$ (nm)	*AB* Magnitude[b]	ID	PI
Cosmic Eye	F606W	**592**	67	20.44	10491	Ebeling
	F814W	800	65	19.89	11160	Richard
	F110W	1123	163	19.97	10491	Ebeling
	F160W	1603	118	19.24	10491	Ebeling
8:00arc	F450W	456	40	20.26	11167	Allam
	F606W	**600**	64	19.69	11167	Allam
	F814W	800	65	19.45	11167	Allam
	F110W	1123	163	19.44	11167	Allam
	F160W	1603	118	19.07	11167	Allam
cB58	F555W	**544**	52	20.70	6832	Yee
	F675W	672	37	20.40	6003	Saglia
	F814W	800	65	20.38	6832	Yee
	F160W	1608	117	19.79	7748	Ellingson
Cosmic Horseshoe	F475W	**477**	42	19.86	11602	Allam
	F606W	589	66	19.63	11602	Allam
	F814W	805	67	19.36	11602	Allam
	F110W	1153	143	18.98	11602	Allam
	F160W	1537	83	18.56	11602	Allam
A68-h7[c]	F814W	**804**	65	21.60	11591	Kneib

[a] The bolded λ_{eff} are the bands used for spatially resolved continuum UV maps to construct SFR maps in Section 6.1

[b] Flux density measurements have $\sim 5\%$ uncertainty due to flux calibration and aperture choice, with comparatively negligible statistical uncertainties.

[c] Due to the inaccessibility of most of the archival optical data for this source, we adopt the photometry from P. Sklias et al. (2014) in order to ensure consistent choices in aperture selection, foreground subtractions, etc. across all bands in the spectrum.

2014 October 17 and 2016 February 25 had to be discarded due to high flagging fractions, and data taken on 2016 April 25 had to be discarded due to missing flux calibration data. The peak signal in the VLA data is $\sim 5\sigma$ with $\sigma = 0.012$ mJy beam^{-1} in the 300 km s^{-1} integrated line map. Including low resolution data from D. A. Riechers et al. (2010) did not improve the imaging result significantly.

We downloaded 2013.1.00139.S ALMA data (PI: N Tohru) from the archive, which contained both 850 μm Band 7 and 650 μm Band 8 data. Dust continuum was detected in the Band 7 data, whereas it was not detected in Band 8. Resolved [C I] ($^2P_{3/2} - ^2P_{1/2}$) lines emission at $\nu_{rest} = 1900.5372$ GHz was detected in the Band 8 data, with a FWHM of ~ 300 km s^{-1}. The FWHM of the high-resolution [C I] detection was used as the line width for the integrated CO(1–0) map. Despite the earlier low-resolution detection in D. A. Riechers et al. (2010) with a FWHM of ~ 200 km s^{-1}, our higher resolution data appears to better match the [C I] line width since it produces a larger SNR. Imaging data obtained from the *HST* archive includes the F606W band (from proposal 10491; PI: Ebeling), and the F814W, F110W, and F160W bands (from proposal 11160; PI: Richard). The foreground lensing galaxy was not detected in the ALMA image. In the *HST* images, the foreground lens and the Cosmic Eye are sufficiently separated for there to be minimal contamination to our measured optical fluxes. There is no radio continuum detected at the 3σ limit in the VLA observations for both the lens and the Cosmic Eye, with $\sigma = 3\,\mu$Jy/bm.

4.2. *8:00arc*

We obtained VLA data from program 18A-114 (PI: C Sharon). 3C48 was used as the flux and bandpass calibrator. J0022+0608 was used as the amplitude and phase calibrator. Data taken on 2018 October 10 was excluded from imaging because of missing flux calibration observations. The data was obtained in the D configuration, and there is significant continuum from the foreground lensing galaxy. The CO(1–0) line was not detected at the 3σ level, where $\sigma = 0.035$ mJy beam^{-1} in the 200 km s^{-1} integrated line map. Robust imaging with a Briggs weighting factor of 0.5 was also performed, but similarly did not yield a line detection.

We obtain ALMA 1.3 mm Band 6 data from archival ALMA projects 2013.1.00262.S and 2015.1.01157.S (PI: S Malhotra) and archival *HST* data using the F450W, F606W, F814W, F110W, and F160W filters from *HST* proposal 11167 (PI: Allam). The *HST* imaging shows additional sources, potentially lensed, in addition to the 8:00arc itself. For this analysis, we use only the extended arc since that represents the majority of emission for our target. The counterimage (which is merged with the lens in the ALMA data) has less than 10% the overall flux of the source, and therefore does not significantly affect the extraction of physical parameters for the source (such as its stellar mass or SFR) cosidering other uncertainties. The lens is bright in the ALMA band, showing two resolved point sources that are also detected in the VLA continuum.

The Band 6 ALMA data also contains a resolved detection of the CO(7–6) line, but offset by 400 km s^{-1} relative to our fiducial redshift (Table 1) and with a much narrower line FWHM of $\sim$ 150–200 km s^{-1}. We therefore performed additional imaging of the CO(1–0) line in the VLA data at the frequencies suggested by the CO(7–6) detection, but the CO(1–0) line remained undetected at the 3σ level.

4.3. *cB58*

cB58 has existing unresolved CO(1–0) line emission obtained from the VLA (D. A. Riechers et al. 2010) as well as CO(3–2) emission obtained from the IRAM Plateau de Bure Interferometer (A. J. Baker et al. 2004). Higher resolution CO(1–0) observations were carried out in the VLA as part of 14B-278 and 16A-077 (PI: C Sharon). J1506+3730 was used as an amplitude and phase calibrator. 3C286 was used as the flux and bandpass calibrator. Due to the proximity and brightness of the foreground lens, CO(1–0) imaging was carried out using a robust weighting with Briggs parameter of 0.5 and the Hogbom CLEAN algorithm. No CO(1–0) detection was made at the 3σ level, with $\sigma = 0.021$ mJy beam^{-1} in the 200 km s^{-1} integrated line map. Imaging was also carried out with natural weighting to verify the non-detection. Including low resolution data from D. A. Riechers et al. (2010) did not improve the imaging result.

ALMA 890 μm Band 7 observations were carried out as part of 2019.1.01148.S (PI: C Sharon). We also examined archival ALMA 1.3 mm Band 6 data from 2013.1.00262.S and 2015.1.01157.S (PI: S Malhotra). While the foreground lens is detected in both bands, the data is of sufficient resolution that the lens does not blend with cB58 when imaged using robust weighting. We also examined archival *HST* imaging from several filters: F160W from proposal number 7748 (PI: Ellingson), F675W from 6003 (PI: Saglia), and both the F555W and F814W filters from 6832 (PI: Yee). The *HST* images show multiple nearby sources in addition to the lens. However, the data are of sufficient resolution that emission from the lens is not expected to significantly affect our optical fluxes for cB58.

4.4. *Cosmic Horseshoe*

The Cosmic Horseshoe was not observed as part of the CO(1–0) observations, but it was observed at 870 μm in ALMA Band 7 as part of 2019.1.01148.S (PI: C Sharon) and 1.3 mm Band 6 in the archival ALMA project 2013.1.00262.S (PI: S. Malhotra). We include it in our analysis since the high-quality ALMA observations allows us to explore our simulated imaging methodology on a very extended source (Section 7). The foreground lens is not detected in either ALMA band at the 3σ limit. The Cosmic Horseshoe itself is not detected in Band 6, nor was the CO(3–2) line. We also analyze archival *HST* imaging using from *HST* Proposal 11602 (PI: Allam) using the F475W, F606W, F814W, F110W, and F160W filters. The lens is UV-bright but does not significantly contaminate our measured *HST* fluxes.

4.5. *A68-h7*

The CO(3–2) line was previously detected for A68-h7 using the 30m IRAM telescope (M. Dessauges-Zavadsky et al. 2015). We obtained VLA observations of the CO(1–0) line as part of 18A-114 (PI: C Sharon). The flux and bandpass calibrator was 3C48. The phase and amplitude calibrator was J0022+0608. The CO(1–0) line was not detected at the 3σ level, where $\sigma = 0.052\,\mathrm{mJy\,beam^{-1}}$ in the $300\,\mathrm{km\,s^{-1}}$ integrated line map.

We also analyse archival *HST* imaging using the filters F132N from proposal ID 12197 (PI: Richard) and F814W from 11591 (PI: Kneib). There is likely some flux contamination from nearby sources whose relation to A68-h7 is unknown. Morphology suggests A68-h7 is a spiral with a relatively weak magnification factor of 2 (M. Dessauges-Zavadsky et al. 2015). P. Sklias et al. (2014) suggest that A68-h7 is an interacting system, with the eastern spiral galaxy and the other nearby systems (located to the northwest, southwest, and more centrally; see Fig. 1) constituting a merger.

5. SPECTRAL ENERGY DISTRIBUTION FITTING

In order to prepare our maps for constraining the Schmidt-Kennicutt relation (Sections 6) and better understand our CO(1–0) non-detections (Section 7), we fit an spectrum to the available photometry in order to constrain the dust luminosity (L_{TIR}, calculated from 3–1100 μm), SFR, and stellar mass ($M_\star$). Spectral fitting was done using Prospector, which is a flexible Markov chain Monte Carlo (MCMC) based spectral fitting tool developed by B. D. Johnson et al. (2021) that has been used on both low-redshift (e.g., J. Leja et al. 2017) and high-redshift (e.g., B. Wang et al. 2023, 2024) galaxies.

We use a non-parametric star-formation history (SFH) with flexible stellar mass formed over five time bins (the "continuity SFH" in Prospector stellar population models). To model the IR emission, we used the built-in dust models (B. T. Draine & A. Li 2007), which has three free parameters: $q_{\rm PAH}$ (the fraction of the dust mass in polycyclic aromatic hydrocarbons; PAHs), $\mathrm{U_{min}}$ (the minimum radiation field intensity that the dust is exposed to, expressed in terms of the Milky Way radiation field), and γ (the fraction of dust heated by starlight with $\mathrm{U} > \mathrm{U_{min}}$). We set uniform priors for all three parameters, with $\gamma \in (0, 1)$, $q_{\rm PAH} \in (0, 10)$, and $\mathrm{U_{min}} \in (0.1, 50)$. We used Levenburg-Marquardt least-squares optimization to initialize our parameters from 50 random draws, and `emcee` was used for the MCMC sampling of the parameter space. We obtain posterior distributions for our parameters, which we then sample to specify the best-fit values and their uncertainties.

In addition to the ALMA, *HST*, SDSS, and *Spitzer* data described in Section 3, we use published photometry from rest-frame UV to the far-IR to constrain the spectrum. Data obtained from the literature and their references can be found in Table 10 and Table 11 in Appendix A. Except for the high-resolution ALMA and *HST* data, there is likely to be flux contamination from nearby sources, particularly the central lens in the case of the four galaxies with bright central lenses. As we do not have lensing magnification values for each band (which could vary from band-to-band due to differential lensing; e.g., R. D. Blandford & R. Narayan 1992; A. W. Blain 1999; S. Serjeant 2012), we fit global flux values; therefore the resulting best-fit parameters are uncorrected for lensing and may have additional systematic uncertainties. Optical data points in this work have been corrected for interstellar dust extinction, including literature values when references explicitly state dust-uncorrected values in their results. Systematic uncertainties in the optical fluxes of A68-h7 are possible as we assume values in the Appendix of P. Sklias et al. (2014) are dust-corrected. Any differences due to this assumption for the optical fluxes are likely not significant considering the systematic uncertainties inherent to photometric fitting and additional uncertainty in identifying the galaxies that comprise A68-h7 in the optical bands (the largest change would be $\sim 25\%$ in the B band).

We present our best-fit values for the SFR, L_{TIR}, and (M_*) in Table 8. The reported values correspond to the median of the posterior distribution. We use the difference between the 75th percentile value and the median value as our upper uncertainty on the reported value,

Table 6. Summary of archival ALMA observations and continuum detections

Source	ν (GHz)	S mJy	σ_{rms} (mJy/beam)	Beam arcsec(″)×arcsec(″)	ID	PI
Cosmic Eye	**340**	2.70 ± 0.35	0.035	0.40×0.34	2013.1.00139.S	N. Tohru
	465	$< 3\sigma$	7	0.33×0.31	2013.1.00139.S	N. Tohru
	95	$< 3\sigma$	0.005	0.97×0.84	2019.1.00790.S	A. Man
					2012.1.00008.S	K. Coppin
					2019.1.01642.S	K. Coppin
8:00arc	**230**	1.91 ± 0.23	0.03	0.89×0.83	2013.1.00262.S	S. Malhotra
					2015.1.01157.S	S. Malhotra
	135	0.30 ± 0.05	0.02	1.88×1.09	2015.1.01157.S	S. Malhotra
	100	$< 3\sigma$	0.004	0.20×0.18	2016.1.01231.S	G. Drouart
					2019.1.00790.S	A. Man
cB58	**230**	0.83 ± 0.18	0.027	1.34×0.80	2013.1.00262.S	S. Malhotra
					2015.1.01157.S	S. Malhotra
	340	1.39 ± 0.29	0.090	1.61×0.91	2019.1.01148.S	C. Sharon
	140	$< 3\sigma$	0.02	1.9×0.94	2015.1.01157.S	S. Malhotra
	108	$< 3\sigma$	0.008	3.94×2.46	2019.1.00790.S	A. Man
Cosmic Horseshoe	**340**	2.80 ± 0.31	0.015	1.07×0.98	2019.1.01148.S	C. Sharon
	230	$< 3\sigma$	0.038	1.58×1.13	2013.1.00262.S	S. Malhotra
	95	$< 3\sigma$	0.015	0.2×0.11	2016.1.01231.S	G. Drouart

NOTE—The bolded frequencies are those that are used for the spatially resolved IR tracer for constructing the SFR maps in Section 6.1.

Table 7. Summary of new VLA CO(1–0) observations

Source	Array	$\nu_{\rm obs}$ (GHz)	S^a (mJy)	σ_{rms} (mJy/beam)	Time (hr)	Beam arcsec(″)×arcsec(″)	ID
Cosmic Eye	C[a]	28.2944	0.245 ± 0.044	0.012	29.7	1.26×1.09	14B-278, 16A-077
8:00arc	D	30.9141	$< 3\sigma$	0.033	5.22	2.77×2.40	18A-114
cB58	C	30.9121	$< 3\sigma$	0.029	14.08	0.91×0.74	14B-278, 16A-077
Cosmic Horseshoe	...	33.1140	...	...	...		
A68-h7	D	36.5604	$< 3\sigma$	0.05	1.30	2.78×2.17	18A-114

[a] Data taken with CnB configuration were uv-clipped to the maxmimum baseline in C configuration.

and the difference between the 25th percentile and the median as our lower uncertainty.

The plots of our best-fit spectra are shown in Appendix A. The best-fit parameters from our spectra are generally consistent with those from A. Saintonge et al. (2013) and P. Sklias et al. (2014) when considering magnification corrections, although systematically lower. Our stellar masses are on average ~ 0.3 dex lower, with a maximum of ~ 0.8 dex for the 8:00arc. Our SFR are on average $\sim 80\%$ of the values, with the lowest being $\sim 75\%$ for the 8:00arc. The $L_{\rm TIR}$ values are broadly consistent, although for the 8:00arc our value is about $\sim 50\%$ compared to that in A. Saintonge et al. (2013). These systematic differences likely occur due to differences in photometry fitting methodology: we jointly fit flux measurements from the optical through the far IR (including new data from ALMA, etc.), so the best-fit model compromises between optimizing fits in the optical/near-IR (which largely sets the stellar mass) and the far IR (which dominates the SFR in these galaxies). Based on the results of the spectral fits and the MS adopted from K. E. Whitaker et al. (2012), we confirm that the galaxies in our sample are MS galaxies. All the galaxies we analyzed were within the 0.34 dex scatter of the MS relation in K. E. Whitaker et al. (2012) except the 8:00arc which has an offset of about 0.48 dex.

Table 8. Best-fit spectrum parameters and other derived values *without* magnification corrections

Source	SFR	$\log(L_{\rm TIR}/L_\odot)$	$\log(M_*/M_\odot)$	$\alpha_{\rm CO}$
	$(10^3\,{\rm M}_\odot\,{\rm yr}^{-1})$			$({\rm M}_\odot\,({\rm K\,km\,s^{-1}\,pc^2})^{-1})$
Cosmic Eye	1.8 ± 0.4	13.11 ± 0.05	$11.89^{+0.23}_{-0.16}$	4.40
8:00arc	3.2 ± 0.2	13.32 ± 0.03	11.19 ± 0.12	7.78
cB58	1.5 ± 0.1	13.00 ± 0.03	11.22 ± 0.22	13.70
Cosmic Horseshoe	0.96 ± 0.24	12.83 ± 0.07	11.74 ± 0.19	13.22
A68-h7	0.84 ± 0.04	12.82 ± 0.01	11.61 ± 0.05	7.54

6. COMPARISONS TO THE SCHMIDT-KENNICUTT RELATION

While our CO(1–0) observations mostly resulted in non-detections, we are still able to improve surface density limits for these galaxies and thus infer where these galaxies would fall on the Schmidt-Kenicutt relation. Here we briefly describe the construction of the resolved SFR map and assumptions used for the gas map upper limits before discussing how these galaxies compare to other high-redshift sources on the Schmidt-Kennicutt relation.

6.1. *SFR surface density*

Since galaxies at high redshift are known to have clumpy UV star-forming regions (e.g., R. Genzel et al. 2011; N. M. Förster Schreiber et al. 2011; F. Bournaud et al. 2014; Y. Zhu et al. 2026) and patchy dust obscuration (e.g., N. A. Reddy et al. 2010, K. L. Adelberger & C. C. Steidel 2000, J. A. Hodge et al. 2016), we combine rest-frame UV data from *HST* and resolved dust continuum imaging from ALMA to construct a SFR map that accounts for both the obscured and unobscured star formation.

For tracing the unobscured SFR, we use data from the *HST* filter closest in proximity to the center of the *GALEX* FUV band in each galaxy's rest frame (in order to most closely match the original hybrid SFR tracer from C.-N. Hao et al. 2011). The *HST* flux at the selected band is then scaled to the spectrum-derived flux at the restframe *GALEX* FUV band by redshifting the filter response for the FUV band by $(1 + z_{source})$ and calculating the weighted sum of the flux density with respect to the filter transmission. The map is then converted to Solar luminosity units using the expression

$$L_{\rm UV} = \frac{4\pi D_L^2 f_\nu \nu}{L_\odot}, \tag{2}$$

where D_L is the luminosity distance, f_ν is the flux density, and ν is the center frequency of the FUV filter.

For tracing the obscured SFR, we scale the ALMA continuum map to the TIR luminosity obtained from the spectral fits in Section 5, adopting the best-fit value in Table 8. Although using continuum maps nearer the peak of the dust spectrum would better trace the total dust luminosity, given the availability of data with the appropriate spatial resolution, we assume the existing ALMA observations further down the Rayleigh-Jeans tail are representative of the TIR emission. While there is evidence for dust temperature gradients within high-redshift galaxies (e.g., T. Tsukui et al. 2023; R. Fernández Aranda et al. 2025), central hot components represent a relatively small fraction of the emission in the ALMA bands used here, and a larger systematic uncertainty is associated with the use of unresolved photometric fitting to obtain the TIR luminosity (e.g., T. Tsukui et al. 2023; J. Li et al. 2024).

In order to combine the unobscured and obscured tracers, we re-grid and smooth the higher resolution *HST*-derived map to match the pixel size and resolution of the ALMA images. We then combine the $L_{\rm UV}$ and $L_{\rm TIR}$ maps to determine the SFR (in units of $M_\odot$ yr^{-1}) using

$$\mathrm{SFR} = 4.6 \times 10^{-44}(L_{\rm UV} + 0.46 L_{\rm TIR}) \tag{3}$$

from C.-N. Hao et al. (2011). However, we only use this expression to appropriately weigh the UV and TIR contributions to the SFR surface density. We then rescale this resulting SFR map to the photometry-derived SFR value since that is the most accurate SFR based on all available data. Typically, the ratio of SFR values before and after scaling are 0.79, with the maximum 1.12 for the Cosmic Horseshoe and minimum 0.45 for A68-h7. This scaling corresponds to a vertical offset on the SK relation of about of ~ 0.1dex (0.34 dex for A68-h7) on the $\log(\Sigma_{\rm SFR})$ axis. For A68-h7, which lacks ALMA data, the UV map is scaled directly to the photometry-derived SFR given in Table 8.

6.2. *Gas mass surface density*

To obtain upper limits on the gas mass surface density from our observations of the CO(1–0) line emission, we first convert our integrated line maps into luminosity units using

$$L'_{\rm CO} = 3.25 \times 10^7 S_{\rm CO}\Delta\nu D_L^2 (1+z)^{-3} \nu_{\rm obs}^{-2} \quad (4)$$

from P. M. Solomon et al. (1992), where $L'_{\rm CO}$ is the line luminosity (in $\rm K\,km\,s^{-1}\,pc^2$), $S_{\rm CO}\Delta\nu$ is the integrated flux (in $\rm Jy\,km\,s^{-1}$), $D_{\rm L}$ is the luminosity distance (in Mpc), and $\nu_{\rm obs}$ is the frequency of observation (in GHz). To convert the line luminosity map to a gas mass, we must assume a CO-to-H_2 conversion factor, $\alpha_{\rm CO}$. We explore a range of possible $\alpha_{\rm CO}$ values, including standard assumptions of a U/LIRG-like value ($\alpha_{\rm CO} = 0.8\,{\rm M}_\odot\,({\rm K\,km\,s^{-1}\,pc^2})^{-1}$) (e.g., N. Z. Scoville et al. 1991; D. Downes & P. M. Solomon 2003) and the Milky Way value ($\alpha_{\rm CO} = 4.36\,{\rm M}_\odot\,({\rm K\,km\,s^{-1}\,pc^2})^{-1}$) (e.g., R. B. Larson 1981; A. W. Strong & J. R. Mattox 1996). However, observational evidence suggests that $\alpha_{\rm CO}$ varies with metallicity and other environmental factors (e.g., A. D. Bolatto et al. 2013; Y.-H. Teng et al. 2023). To account for some of this variation, we also use the metallicity and redshift-dependent $\alpha_{\rm CO}$ from G. Accurso et al. (2017) ($\alpha_{\rm CO}^{\rm Acc.}$, hereafter) which also includes a small correction for offsets from the galaxy MS. Following G. Accurso et al. (2017), we adopt the form of the MS presented in K. E. Whitaker et al. (2012):

$$\log({\rm sSFR}/[{\rm Gyr}^{-1}]) = -1.12 + 1.14{\rm z} - 0.19{\rm z}^2 - (0.3 + 0.13z)(\log({\rm M}_*) - 10.5) \quad (5)$$

where $\rm M_*$ is the stellar mass of the galaxy (in units of $\rm M_\odot$), and $\rm sSFR = SFR\,M_*^{-1}$ is the specific-SFR (in units of $\rm Gyr^{-1}$). We use the stellar masses and the SFR from our spectral fits (as shown in Table 8) which are then corrected for the magnification factors (Table 1). The metallicities for most galaxies were obtained from A. Saintonge et al. (2013) and references therein. A68-h7 does not have a reported metallicity; instead, we use the average metallicity for unlensed galaxies (since A68-h7 is weakly lensed) observed in the appropriate redshift and stellar mass range from T.-T. Yuan et al. (2013).

6.3. *The SK Relation*

We show the resulting SK relation for our observations in Fig. 2. Since most of the CO(1–0) observations resulted in non-detections, we place galaxies on the SK relation using the average $\Sigma_{\rm SFR}$ within its 2σ detection boundary and the 3σ upper limit from $\Sigma_{\rm gas}$. Since the Cosmic Eye CO(1–0) detection was marginally resolved (with emission spread over ~ 2 beams), for that source we computed the surface density where the signal in the CO(1–0) integrated line image was above 2σ. In order to assess how our high-resolution upper-limits compare to previous CO detections of these sources, we also show an average gas mass surface brightness using integrated CO(3–2) detections from the literature and the 2σ $\Sigma_{\rm SFR}$ boundary for the source size. We note that no flux correction factors to account for emission beyond the 2σ boundary were applied for any of our sources since the additional emission was found to be negligible (i.e. comparable to the noise in the image).

In general, we find that our high resolution CO(1–0) observations are offset to lower gas mass surface brightnesses than for the CO(3–2) detections, particularly when accounting for the fact that CO(1–0) observations are mostly upper limits. The comparison to the CO(3–2) data is somewhat complicated by the necessary assumption of a CO(3–2)/CO(1–0) line ratio ($r_{3,1} = L'_{\rm CO(3-2)}/L'_{\rm CO(1-0)}$) to correct for gas excitation conditions. In Fig. 2, we show the range in possible average gas mas surface densities for different line ratios when using the CO(3–2) line. For the Cosmic Eye (our marginal detection) and the 8:00arc (a 3σ upper limit), the CO(3–2) values are barely consistent with the assumption of a thermalized line ratio ($r_{3,1} = 1$). This result is unsurprising given the weak/noisy detection with a somewhat lower total flux than the previous low-resolution CO(1–0) observation that yielded an observed line ratio of $r_{3,1} = 0.72 \pm 0.16$ (D. A. Riechers et al. 2010). The 8:00arc did not have a prior unresolved CO(1–0) detection, so our upper limit suggests that thermalized line ratios are more appropriate for this source. For A68-h7, our 3σ upper limit is well below the range in average surface densities for the CO(3–2) observation. This apparent discrepancy is most likely due to the ambiguity in the source size for A68-h7; in *HST* images, A68-h7 has four components in close proximity (see Fig. 1), and it is unclear which components contribute to the existing CO(3–2) detection. In addition, we do not have resolved dust continuum maps for A68-h7, and there could be extended emission associated with dust obscured and optically dark star formation that is not currently accounted for (as seen for cB58). Redshifts of the individual components and/or higher resolution CO(3–2) observations are necessary to better constrain the emission source and thus surface area. Only the CO(1–0) 3σ upper limit for cB58 is consistent with any possible value of $r_{3,1}$ for the CO(3–2) average surface brightness, due to the low depth of the CO(1–0) observations. At the time of the CO(1–0) observations, the

ALMA dust emission map did not exist, so observations were informed by the much more compact (rest-frame) UV emission in the *HST* data.

Even though most of the CO(1–0) observations resulted in upper limits, these results strongly suggest that U/LIRG-like assumptions of $\alpha_{\rm CO}$ are inappropriate for high-redshift MS galaxies (with the possible exception of cB58). U/LIRG-like $\alpha_{\rm CO}$ values place these systems at higher SFEs than seen in any other high-redshift system, including starbursts like GN20 (J. A. Hodge et al. 2015) and AzTEC-1 (K. Tadaki et al. 2018). These galaxies would shift to even higher SFEs when accounting for the fact that the data are (largely) upper limits on the gas. The range of possible values based on the disk-averaged CO(3–2) detections also reinforces this conclusion. Choices in $\alpha_{\rm CO}$ informed by metallicity values or dynamical constraints are therefore critical for avoiding non-detections of resolved gas observations in future observations of high-redshift MS galaxies.

Previous work has suggested that the offset SK relations between starburst galaxies (like SMGs and U/LIRGs) and more "normal"/MS disk-like galaxies (e.g., E. Daddi et al. 2010) is largely due to the bimodal assumption in $\alpha_{\rm CO}$ values (e.g., R. Genzel et al. 2010; R. C. Kennicutt & M. A. C. De Los Reyes 2021). However, our upper limits and the marginal detection for the Cosmic Eye suggest that even with larger values of $\alpha_{\rm CO}$, similar to the Milky Way value that corresponds to the "sequence of disks," some galaxies genuinely have higher SFEs that place them nearer to the "sequence of starbursts." Therefore, choice in $\alpha_{\rm CO}$ is not solely/artificially responsible for some galaxies being offset to higher SFEs in the SK relation (see also: C. E. Sharon et al. 2013, 2019; R. C. Kennicutt & M. A. C. De Los Reyes 2021). A more uniform analysis of the SK relation with continuous forms of the CO-to-H_2 conversion factor is necessary to understand the true scatter in the SK relation, and what role (if any) galaxy type or redshift may play in that scatter.

A potential caveat to these conclusions regarding $\alpha_{\rm CO}$ values and the offset of galaxies to higher SFEs on the SK relation is due to the uncertainty in the resolved SFRs, particularly the obscured star formation. As described in Section 6.1, we scale the high-resolution ALMA data to the $L_{\rm TIR}$ inferred from the spectral fits. The IR peak in the spectrum is largely dominated by low-resolution *Herschel* data, which can cause some issues in the photometric fits. For the 8:00arc and cB58 specifically, there are nearby IR-bright galaxies that may be blended with our target and therefore significantly contribute to the *Herschel* fluxes. Thus it is possible that these two sources should have lower overall SFRs, which moves them closer to the "sequence of disks" in the SK relation. A resolved IR map nearer the dust peak and existing *Herschel* detections is important for determining how much contamination from nearby sources is contributing to $L_{\rm TIR}$ for these galaxies. In addition, for all strongly lensed sources, differential lensing could be biasing the overall photometry to more bright/compact features within the galaxies, resulting in spectral fits that are not representative of the whole galaxy (e.g., Y. D. Hezaveh et al. 2012; S. Serjeant 2012).

To better constrain possible $\alpha_{\rm CO}$ values and possible forms of the SK relation, as well as to understand our non-detections, we simulate resolved CO(1–0) VLA observations using the SFR maps in the next section.

7. CONSTRAINTS FROM SIMULATED CO(1–0) MAPS

7.1. *Constructing the Theoretical Resolved Gas Maps*

In order to constrain possible forms for the SK relation and $\alpha_{\rm CO}$ values for our sources, we produce simulated observations of resolved gas maps and compare those results to our observed upper limits. To do this, we assume a range of potential SK relations, and apply those to the SFR maps made in Section 6.1 to construct a theoretical gas map.

We use the SK relations from E. Daddi et al. (2010) for "normal"/disk-like galaxies and starburst galaxies to compute overall relationship normalizations, since those reasonably span the potential range of SFEs observed in previous work. Since there appears to be some variation in resolved galaxies' SK slopes at low redshifts even though the population as a whole follows a single average relation (A. K. Leroy et al. 2013), we also explore different slopes for the SK relation. However, the precise numerical value for the offset, A, in the SK relation (Equation 1) depends on the assumption of the slope, n. So in order to keep the simulated gas maps' average surface brightnesses along one of the E. Daddi et al. (2010) relations, we convert the offset value using the expression

$$A_{\rm mod} = \log\langle\Sigma_{\rm SFR}\rangle(1 - \frac{n_{\rm mod}}{n_{\rm orig}}) + A_{\rm orig}\frac{n_{\rm mod}}{n_{\rm orig}}, \quad (6)$$

where $\langle\Sigma_{\rm SFR}\rangle$ is the average SFR surface density within the 2σ borders of the source, $A_{\rm orig}$ and $n_{\rm orig}$ are for the original form of the SK relation (from E. Daddi et al. 2010 in our case), $n_{\rm mod}$ is the modified slope we want to model, and $A_{\rm mod}$ is the corresponding modified normalization we would need to assume.

We chose $n_{\rm mod}$ to be 1, 1.5, and 2, since those span the typical range of slopes found in observational studies

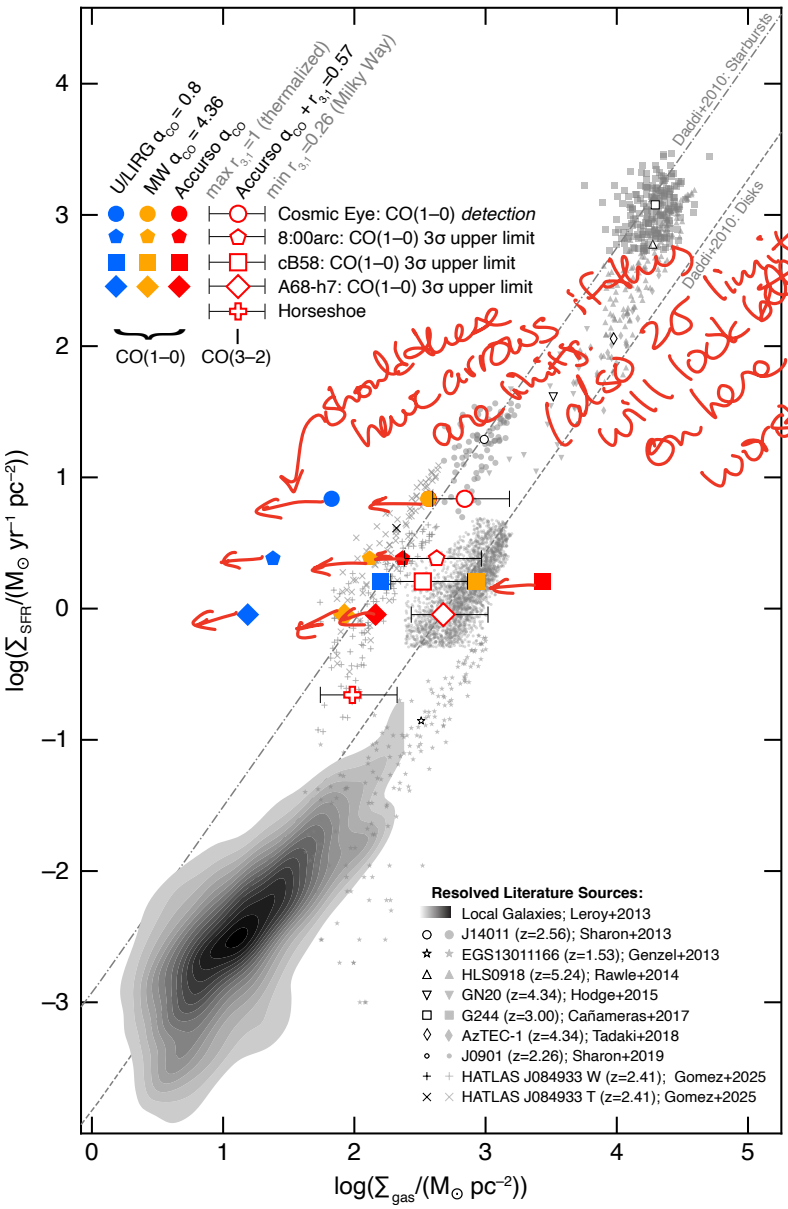


Figure 2. Our sample of galaxies with existing VLA observations plotted on the SK relation (colored points) along with resolved high-redshift galaxies with similar pixel-by-pixel analyses from literature (various gray points, described further below) and a sample of low-redshift galaxies from A. K. Leroy et al. (2013) (grayscale density contours). The blue, gold, and red symbols correspond to different assumptions of α_{CO} used to calculated the gas mass surface densities: the typical U/LIRG value of $0.8\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ (blue), the Milky Way value of $4.36\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ (gold), and the G. Accurso et al. 2017 derived metallicity-dependent value (red). Different symbols correspond to different objects, where the pentagons, squares, and diamonds correspond to 3σ upper limits of CO(1-0) for the 8:00arc, cB58, and A68-h7 respectively, and the circles correspond to the detected CO(1-0) emission for the Cosmic Eye. The Cosmic Horseshoe (plus symbol), was not observed in CO(1–0), but was included as it had high SNR resolved UV and IR detections. The hollow symbols use the CO(3–2) detections to calculate the gas mass surface densities using the G. Accurso et al. 2017 α_{CO} and a standard assumption for the CO(3–2)/CO(1–0) line ratio of $r_{3,1} = 0.72$ (the average value for lensed MS galaxies from A. Saintonge et al. 2013); the horizontal error bars on the CO(3–2) points captures the range in possible CO(3–2)/CO(1–0) line ratios (from the Milky Way value of $r_{3,1} = 0.26$ from D. J. Fixsen et al. (1999), to a thermalized line ratio of $r_{3,1} = 1$). Small gray points are individual surface brightness measurements for high-redshift sources with resolved CO detections, and their average surface brightness is given by the black-outlined point: large circles are from C. E. Sharon et al. (2013), stars are from R. Genzel et al. (2013), upward points triangles are from T. D. Rawle et al. (2014), downward pointing triangles are from J. A. Hodge et al. (2015), squares are from R. Cañameras et al. (2017), diamonds are from K. Tadaki et al. (2018), small circles are from C. E. Sharon et al. (2019), and crosses and exes are from J. S. Gómez et al. (2025). The black dashed line is the SK relation for the "sequence of disks", and the grey dotted-dashed line is the SK relation for the "sequence of starbursts", both from E. Daddi et al. (2010).

(A. K. Leroy et al. 2013; R. C. Kennicutt & M. A. C. De Los Reyes 2021; J. A. Hodge et al. 2016; Y.-C. Chen et al. 2024, e.g.,). The different values of $A_{\rm mod}$ and $n_{\rm mod}$ then allow us to invert the SK relation to obtain theoretical gas maps from our SFR maps, which we can then use as input models for simulated observations of CO(1–0).

7.2. *Simulating Observations of CO(1–0)*

Given our theoretical gas maps for various assumptions of the SK relation, we can then produce simulated observations of the CO(1–0) line that match our observations. As specified previously, the total gas mass of a galaxy is given by $M_{gas} = L'_{\rm CO}\alpha_{\rm CO}$. The gas mass surface density, Σ_{gas}, can be written as

$$\Sigma_{gas} = M_{\rm gas} N \theta(z)^{-2}, \tag{7}$$

where $\Sigma_{\rm gas}$ is the gas mass surface density in units of $M_\odot\,{\rm pc}^{-2}$, $M_{\rm gas}$ is in units of Solar masses, N is the number of pixels in 1 arcsec^2, and $\theta(z)$ is the angular size (in arcseconds) of 1 pc at redshift z. We substitute in the expression for M_{gas} in to obtain:

$$\Sigma_{gas} = \alpha_{\rm CO} L'_{\rm CO} N \theta(z)^{-2}. \tag{8}$$

We then express the line luminosity ($L'_{\rm CO}$) in as a function of the integrated line flux ($S_{\rm CO}\Delta\nu$) using Equation 4, to obtain:

$$\Sigma_{gas} = 3.25\times10^7 \alpha_{\rm CO} S_{\rm CO}\Delta\nu \\ D_L^2 \nu_{obs}^{-2} (1+z)^{-3} \theta(z)^{-2} A_{bm}^{-1}, \tag{9}$$

where $S_{\rm CO}\Delta\nu$ is in units of Jy km s^{-1}, A_{bm} is the area of the beam in arcsec^2, and the other terms are as defined previously. We solve this expression for $S_{CO}\Delta\nu$ and apply it to our theoretical gas maps ($\Sigma_{\rm gas}$) to obtain a predicted CO(1–0) integrated line map. The line width, $\Delta\nu$, is obtained from previous integrated detections of the CO line. In addition to varying the assumed form of the SK relation, we also explored three different $\alpha_{\rm CO}$ values: the $\alpha_{\rm CO}$ derived from using G. Accurso et al. (2017), $\alpha_{\rm CO} = 4.36\,{\rm M}_\odot\,({\rm K\,km\,s^{-1}\,pc^2})^{-1}$ for the Milky Way (P. M. Solomon et al. 1987), and $\alpha_{\rm CO} = 0.8\,{\rm M}_\odot\,({\rm K\,km\,s^{-1}\,pc^2})^{-1}$ for local U/LIRGs (D. Downes & P. M. Solomon 1998).

We then use these theoretical CO(1–0) line maps as input for the simulated observations using `simobserve` in the modular version of CASA 6.7.0. We crop the input maps to prevent flux contamination from secondary sources (such as the foreground lenses) and noise structures from the SFR maps. In order to compare the simulated maps to our data, for each source we match the observing parameters in `simobserve` to our observations as closely as possible: using the same start and end times of each track, the same dates, and the same array configurations.[7] As the Cosmic Horseshoe is a source with (i) existing highly resolved detections in *HST* and ALMA observations and (ii) no prior VLA observations, we simulate its CO(1–0) emission to illustrate how different assumptions about the underlying SK relation affects observability. We use the observational parameters of the Cosmic Eye (which has the longest integration time among our sources) and assume that the Cosmic Horseshoe is at the local meridian in the middle of each observation run, thereby reaching a similarly low $\sigma_{\rm rms}$ to that of the Cosmic Eye.

We present an overview of our simulation results in Table 9, including details about the varying maximum SNRs and integrated fluxes across the six models. We assume slopes for the SK relation of $n = 1$, 1.5, and 2, with normalizations that either place the galaxies' average $\Sigma_{\rm SFR}$ either along the "sequence of disks" or along the "sequence of starbursts" (E. Daddi et al. 2010). In order to better replicate our weak and non-detections, we use only $\alpha_{\rm CO}^{\rm Acc}$ since it produces the dimmest predicted CO(1–0) line emission. We measure the SNR as the ratio of the peak value within the source region to the $\sigma_{\rm rms}$ of the image. In the case we don't detect significant emission in the simulated image, we report the peak SNR within the region with significant emission in the SFR tracer. We do not report integrated fluxes for the case where the SNR < 4.

7.3. *Results from Simulations*

7.3.1. *Simulation comparisons do not constrain the slope of the SK relation*

We find that the choice of the SK relation slope, n, does not significantly affect the signal in the simulated CO(1–0) line maps. As seen in Table 9, within each class of assumed SK relations (the E. Daddi et al. (2010) sequence of disks vs. starbursts), the integrated line flux

[7] We note that prior to imaging our simulated observations, it is possible to introduce velocity structures by assigning different `clean` model components to different frequency channels. Such velocity modeling could be useful for understanding complicated spectral line shapes in sources with spatially unresolved line emission that have much higher angular resolution continuum observations, for example. However, the quality of the current CO(1–0) observations (and their prior unresolved CO detections in the literature) could not constrain predictions from velocity gradient models for this sample of galaxies, and therefore is not explored in this work.

Table 9. Results of simulated CO observations assuming α_{CO}^{Acc}

Sources	Measurement[a]	Disk Models			Starburst Models		
		n = 1	n = 1.5	n = 2	n = 1	n = 1.5	n = 2
Cosmic Eye	$S\Delta\nu$	0.315	0.318	0.309	0.105	0.066	0.096
$\sigma = 0.016$ Jy km/s	SNR	14.51	17.13	13.52	5.76	4.72	5
8:00arc	$S\Delta\nu$	0.512	0.512	0.486	0.14	0.124	0.152
$\sigma = 0.020$ Jy km/s	SNR	24.9	23.4	22.5	6.6	6.5	7.5
cB58	$S\Delta\nu$	0.14	...	...	...	...	...
$\sigma = 0.018$ Jy km/s	SNR	4.4	3.43	3.75	2.4	3.64	2.95
Cosmic Horseshoe	$S\Delta\nu$	0.243	0.216	0.207	...	...	...
$\sigma = 0.015$ Jy km/s	SNR	10.84	8.00	6.93	3.83	3.08	3.86
A68-h7	$S\Delta\nu$	0.192	0.252	0.135	...	...	...
$\sigma = 0.045$ Jy km/s	SNR	7.08	6.71	4.91	2.98	3.39	2.65

[a] $S\Delta\nu$ is calculated using an aperture that includes the entire source, in units of Jy km s^{-1}. The noise values given in the first column are the uncertainties associated with the aperture flux in order to give a better sense of the significance of the integrated line detection. The reported SNR values are the peak pixel value in the aperture divided by the σ_{rms} noise for each image (see Table 7).

varies by $\lesssim 20\%$ with a max of $\sim 50\%$ for the lowest SNR detections. Similarly, the peak SNR in the simulated maps varies by ~ 1–2σ. As an example of the low variation between different assumptions of the SK relation slope, in Figure 3 we show images of the strongest simulated detection in our sample: the 8:00arc.

The apparent lack of effect is due to the limited range in surface density values above our noise levels when evaluating a log-log relationship. Because of the low dependence of the inferred CO(1–0) line strength for the value of the SK relation slope at the significance level of our detections, we conduct all further analyses using the $n = 1.5$ case.

7.3.2. *Explaining CO(1–0) non-detections*

Based on the comparison between our observations and the simulated line maps, our weak and non-detections have several different causes.

For the Cosmic Eye and the 8:00arc, all simulations result in more significant detections than obtained by our observations, even though our results are generally consistent with previous CO(3–2) detections (assuming thermalized line ratios). Using assumptions of a more starburst-like normalization of the SK relation with metallicity-derived α_{CO} generally produce lower SNR detections, and therefore more closely match our observations than alternative assumptions. If we compare the predicted integrated line fluxes to the CO(3–2) detections, models that use the E. Daddi et al. (2010) starburst normalization yields line ratios of $r_{3,1} \sim 0.7$ while the disk-like normalization yields $r_{3,1} \lesssim 0.2$. While low $r_{3,1}$ values are not unphysical, they are not commonly observed in gas-rich high-z galaxies (A. Kirkpatrick et al. 2019), further suggesting that the assumption of the starburst-like normalization is more accurate for these sources. Using our current inferred SFR maps, reducing the simulated CO(1–0) emission further requires that either these sources should be offset to even higher SFEs than typically seen in starburst-like galaxies, or they posses even larger values of α_{CO} than we calculated using G. Accurso et al. (2017). Given observations of other high-redshift sources, uncertainty in α_{CO} is the more likely of these two explanations. One potential source of uncertainty for the derived CO-to-H_2 conversion factors is that the metallicities for these sources were derived using slit spectroscopic line ratios. Since several of these sources present as arcs, only fractions of the sources were covered in the slits, which could lead to biased metallicity calculations, as metallicity gradients have been observed in high-z galaxies (e.g., P. Sharda et al. 2021; Z. Li et al. 2025). Alternatively, the input SFRs may be overestimated by the spectral fits. As discussed in Section 5, blending with nearby sources in unresolved infrared detections and the effects of differential lensing potentially bias the photometry-inferred SFRs to larger values. Barring improved understanding of α_{CO}, higher resolution maps at frequencies nearer the peak of the dust emission spectrum and multi-wavelength lens modeling is necessary to determine the importance of these two effects.

For cB58 and A68-h7, the non-detections are likely caused by inadequate integration time. At the time of the CO(1–0) observations, the surface brightnesses for these objects were poorly understood. cB58's more re-

yes!

→ proper error bars need to be included or it's difficult to conclude anything robustly here (even on $r_{3,1}$, eta...)

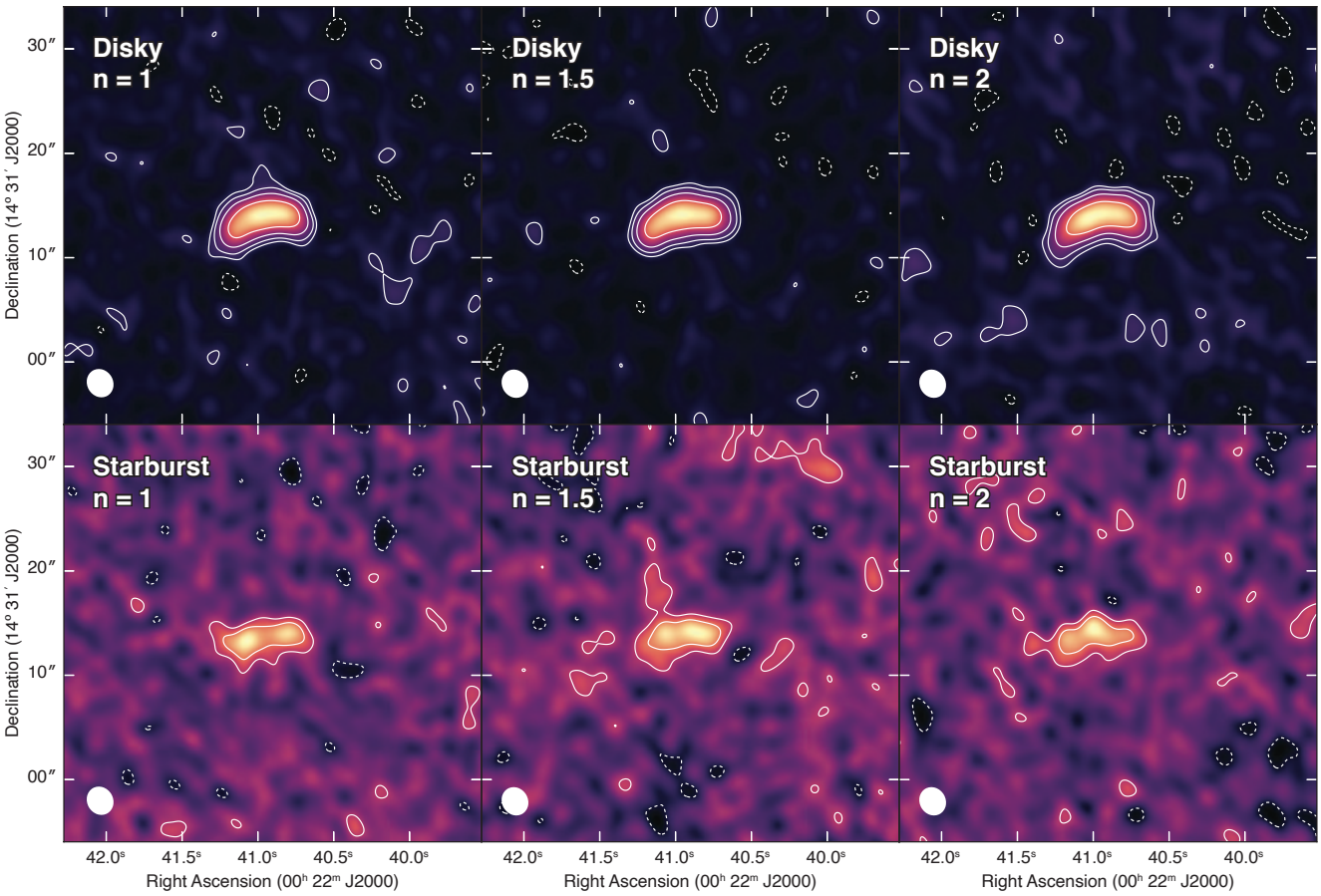


Figure 3. Simulated CO(1–0) integrated line maps with $n = 1$ (left), 1.5 (center), and 2 (right) for the 8:00arc assuming both disk-like (top) and starburst-like (bottom) normalizations of the SK relation from E. Daddi et al. (2010). Contours are powers of $\pm 2 \times \sigma$ (i.e. $\pm 2\sigma$, $\pm 4\sigma$, $\pm 8\sigma$, etc.) for the disk-like normalizations and multiples $\pm 2\sigma$ for the starburst normalizations, where $\sigma = 35\,\mu\mathrm{Jy\,beam^{-1}}$ and negative contours are dashed. The common synthesized beam is located in the bottom left corners.

cent ALMA dust continuum maps show that it is likely more extended than the older *HST* rest-frame UV maps suggest. Since the simulated maps for cB58 produce a $\sim$ 3–4σ peak at best, and our observed CO(1–0) upper limits are above the values inferred from previous CO(3–2) integrated line detections, inadequate integration time is the most straightforward explanation. For A68-h7, as discussed previously, it is unclear which component(s) is (are) primarily responsible for the cold dust and gas emission in the unresolved CO detection from M. Dessauges-Zavadsky et al. (2015). The simulations produce $\sim$ 5–7σ peaks for disk-like assumptions of SK normalization, and weaker $\sim 3\sigma$ emission for the starburst normalization, which suggests additional observations would potentially yield a CO(1–0) detection. While the observed gas surface brightnesses for the two lines in Figure 2 appear inconsistent, that may be a product of using the 2σ boundary from the SFR map when calculating the average gas surface brightness for the integrated CO(3–2) line. In addition to the uncertainty of which optical components are physically associated with the source, A68-h7 also lacks a high resolution dust emission map (similar to cB58 at the time of the VLA observations), making our constraints on the spatial extent (and therefore average surface brightness) even more uncertain. Consequently, inadequate integration times from uncertain surface brightness constraints are the likely cause of the non-detections of the CO(1–0) emission in both cB58 and A68-h7.

While CO(1–0) has not been observed for the Cosmic Horseshoe, it was included in our study due to its resolved high SNR detection with both ALMA and *HST*, and thus its potential for resolved SK studies. From our simulation results, we find that assuming a SK normalization along the "sequence of disks" results in a marginally resolved CO(1–0) line detection in D configuration after 30 hours of integration time. However, assuming a SK normalization along the "sequence of starbursts", the Horseshoe is not detected with the same long integration time. While the Horseshoe is a "nor-

mal" MS galaxy with a larger α_{CO} value, which is suggestive of it falling along the "sequence of disks", that may not be the case (as illustrated by other galaxies in this work: the 8:00arc and Cosmic Eye). These results illustrate the challenges associated with making informed assumptions when attempting resolved molecular gas observations at high redshift.

In short, a combination of shallow observations, uncertain metallicities, overestimated SFRs, and differential lensing likely explain our non-detections, with the exact combination of factors varying from source-to-source. The high-z galaxies in our sample further suggest that some high-redshift MS galaxies may be offset to higher SFE normalizations of the SK relation. Follow-up IFU spectroscopy of the two arcs would improve metallicity constraints and therefore the CO-to-H_2 conversion factors necessary to estimate mapping depths. High frequency ALMA observations that sample the dust spectrum closer to their peaks would also help constrain blending between nearby sources and thus improve SFR estimates.

7.3.3. *Assumption of α_{CO} insufficient to explain offset SK relations*

Previous work has suggested that the offset in the SK relation between starbust galaxies and more "normal" disk-like/MS galaxies may be explained by different assumptions in α_{CO} (e.g., E. Daddi et al. 2010; R. Genzel et al. 2010; R. C. Kennicutt & M. A. C. De Los Reyes 2021). The (in)ability of our simulations to reproduce the weak and non-detections of the Cosmic Eye and 8:00arc (respectively) reinforces our conclusions from Section 6.3 that the choice of α_{CO} is not the sole cause of an offset SK relation to higher SFEs for all galaxies. In Fig. 4, we use the metallicity-dependent form of α_{CO} from G. Accurso et al. (2017), which yields values of the CO-to-H_2 conversion factor that are similar to or larger than the typical Milky Way assumption that ostensibly produces the "sequence of disks" in the SK relation ($\alpha_{CO} = 4.40\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ for the Cosmic Eye and $7.78\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ for the 8:00arc vs. the canonical value of $4.36\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ for the Milky Way; Table 8). However, assuming the disk-like normalization of the SK relation with these larger α_{CO} values produces much more CO(1–0) emission than observed. Instead, the simulations that use "the sequence of starbursts" is a better match to our observations (although they too produce more emission than observed). For the disk-like SK relation to reproduce our non-detections in the simulations would require α_{CO} values that are at least a few times larger than used here, which could occur if these galaxies' metallicites are lower than slit spectroscopy suggests. Therefore, barring improved metallicity measurements, our simulations suggest that both a large CO-to-H_2 conversion factor and a higher SFE form of the SK relation are necessary to reproduce the weak/non-detections of the Cosmic Eye and 8:00arc, and thus the high SFE "sequence of starburst" in the SK relation is not solely due to assumptions of low U/LIRG-like values of $\alpha_{CO} \sim 1\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$.

8. SUMMARY & CONCLUSIONS

In this work, we describe VLA CO(1–0) observations of four $z \sim 2$–3 lensed MS galaxies with prior CO detections (either at lower angular resolution or with higher-J CO lines) and well-constrained IR photometry. The observations resulted in non-detections in three cases (the 8:00arc, cB58, and A68-h7), and a marginal detection in the fourth (the Cosmic Eye). Using new spectral fits to more extensive photometry (Section 5), archival data from ALMA and *HST* to create SFR maps that capture both dust-obscured and unobscured star formation (Section 6.1), and simulated observations (Section 7), we are able to explain the CO(1–0) non-detections and constrain the SK relation for these sources. Our major conclusions are as follows:

- In Section 6, we find that these galaxies' CO(1–0) upper limits and resolved SFRs are more consistent with the "sequences of starbursts" than the "sequence of disks" in E. Daddi et al. (2010) for the SK relation. This result, however, is *not* due to the assumption of a low U/LIRG-like CO-to-H_2 conversion factor; we use a metallicity-dependent form of the conversion factor from G. Accurso et al. (2017) that yields values of $\alpha_{CO} \sim 4$–$14\,M_\odot\,(K\,km\,s^{-1}\,pc^2)^{-1}$ for these galaxies. This result suggests that not all high-redshift galaxies are artificially offset to high SFEs due to inappropriately low assumptions of the CO-to-H_2 conversion factor, and instead some galaxies genuinely have higher SFEs.
- The CO(1–0) observations have mixed consistency with prior CO observations. For the Cosmic Eye, our CO(1–0) detection is broadly consistent with the prior CO(3–2) detection, although we suggest a more thermalized value for the $r_{3,1}$ ratio than inferred from the existing lower resolution CO(1–0) detection (D. A. Riechers et al. 2010). The upper limit of CO(1–0) for cB58 is consistent with the existing CO(3–2) and low resolution CO(1–0) detections. However, the upper limits in A68-h7 and the 8:00arc suggest much lower gas masses than inferred from the prior CO(3–2) detections. These differences could be due to the low SNR of these

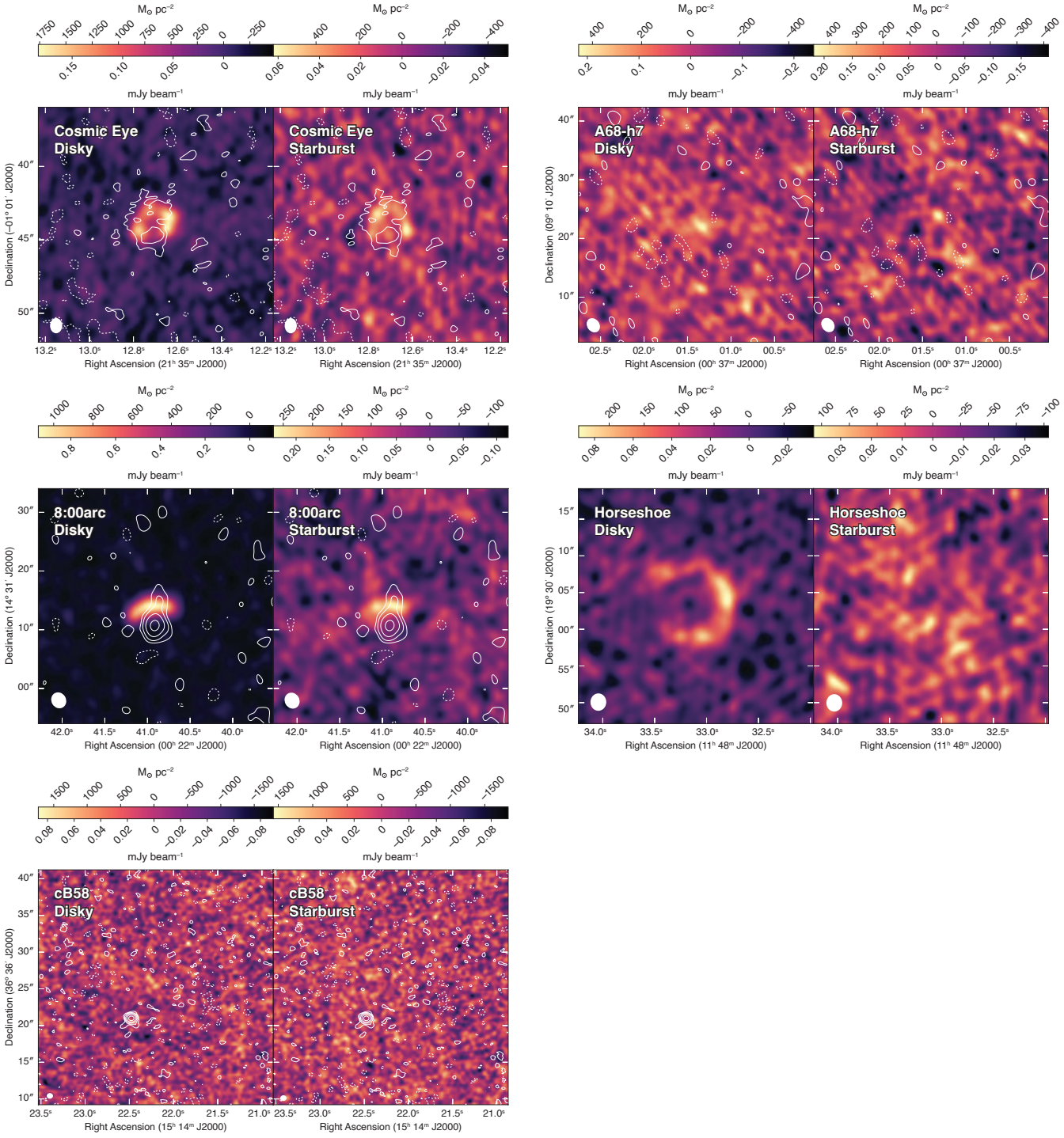


Figure 4. Simulated CO(1–0) observations (image) vs. our actual observations (contours) for the four sources with existing VLA observations plus the Cosmic Horseshoe. Each source was simulated assuming the SK normalization of the "sequence of disks" (left panels) and "sequence of starbursts" (right panels) from E. Daddi et al. (2010). The synthesized beams for the simulations (which match the observations) are located in the bottom left of each panel. The contours are are multiples of $\pm 2\sigma$, and negative contours are dashed; values for σ match those in Figure 1. The scale of the colorbars are given at the top of each panel, in observational units (mJy beam^{-1}) as well their corresponding $\Sigma_{\rm gas}$ units ($M_\odot$ pc^{-2}).

detections, as well as our relatively shallow observations (in the case of A68-h7). Furthermore, it is unclear which components of A68-h7 contribute to observed emission at longer wavelengths. Follow-up higher resolution observations of the CO(3–2) line could help determine which spatial components contribute to the CO emission.

- We compared our observations to simulated observations of the CO(1–0) emission produced from our SFR maps and several different assumptions for the SK relation. We found negligible variation in the simulated maps of the CO(1–0) line emission when using different slopes of the SK relation (specifically slopes of $n = 1$, 1.5, and 2) both qualitatively and quantitatively (as probed by the peak SNRs and integrated line strengths). Therefore we cannot constrain the slope of the SK relation for these galaxies with our current data.

- These simulations suggest that the non-detections of the CO(1–0) line for cB58 and A68-h7 were due to inadequate integrations time since most simulations matched to our observations were able to produce a detectable signal. For the Cosmic Eye and the 8:00arc, however, all simulations produced a more significant detection than we observed. Simulated maps produced using higher values of the CO-to-H_2 conversion factor (in line with these galaxies' low metallicities) and higher SFE normalizations for the SK relation (similar to what is seen for local U/LIRGs) are required to better match our observations. This result further reinforces our prior conclusion that some galaxies have genuinely higher SFEs, and not all high SFEs at high redshift are due to low assumptions of α_{CO}.

The combination of our simulation results and the prevalence of non-detections in our sample suggests that spatially resolved observations of CO lines in high-redshift galaxies requires significant care given the large investments of telescope time involved. Non-detections of CO in the literature and telescope archives, some of which show galaxies similarly offset from the SK relation (for example in R. Decarli et al. (2016), although shown in a luminosity-luminosity plot) suggest more careful derivations of constraints for Σ_{CO} is required. Archival VLA and ALMA non-detections of CO lines may therefore be the result of poorly constrained values of α_{CO} for these sources (due to missing metallicity measurements) and/or uncertain spatial extent of the emission (due to maps that miss either the obscured or unobscured star formation). Along those same lines, the two most significant sources of uncertainty in our conclusions are (a) uncertain slit spectroscopy derived metallicities of these extended objects (necessary for determining α_{CO}) and (b) potential blending with nearby sources near the peak of the dust spectrum (which dominates the SFRs for these sources). In order to understand how star formation processes might change with redshift and galaxy type using the spatially resolved SK relation therefore requires a carefully constructed survey with ample multi-wavelength ancillary data.

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Software: Astropy (Astropy Collaboration et al. 2022), CASA (CASA Team et al. 2022)

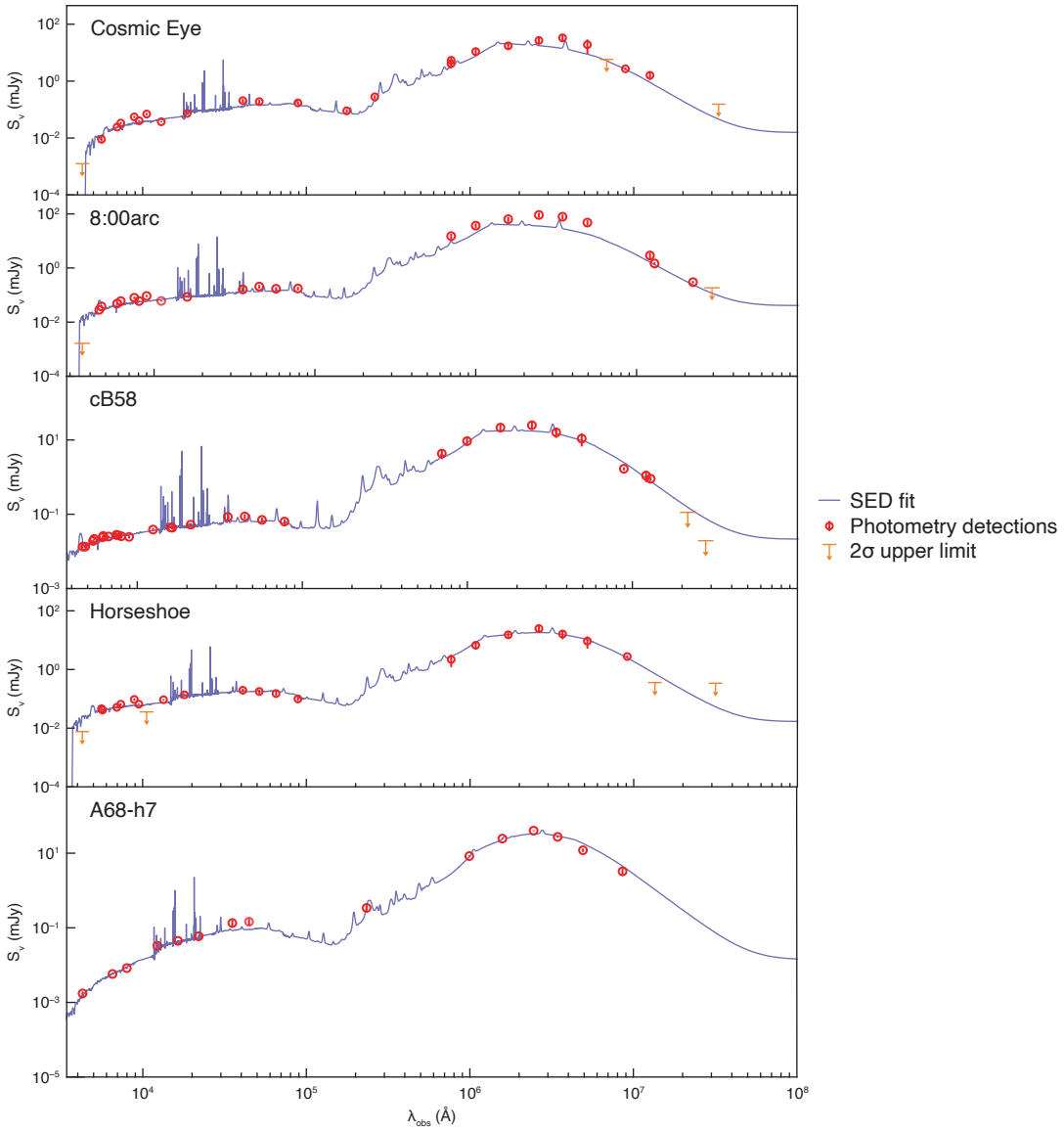


Figure 5. Best-fit spectrum (blue lines) to the observed and new photometry (red points); 2σ upper limits for non-detections are shown with the orange downward-pointing arrows. Best-fit values for the model parameters are given in Table 8.

APPENDIX

A. PHOTOMETRIC DATA AND BEST FIT MODELS

Here we show the best fit spectra in Figure 5 and the input data in Tables 10 and 11, with citations. The description of the photometric fitting methodology is given in Section 5 and resulting best-fit values for the parameters of interest are in Table 8.

Table 10. Optical photometry used for spectrum fitting

		Cosmic Eye		8:00arc		cB58		Cosmic Horseshoe		A68-h7	
Filter	Wavelength	Flux	Ref.	Flux	Ref.	Flux	Ref.	Flux	Ref.	Flux	Ref.
	(μm)	(AB Mag)		(AB Mag)		(AB Mag)		(AB Mag)		(AB Mag)	
SDSS u	0.35	< 24.40	This Work	< 24.12	This Work			< 22.44	This Work		
CFHT 12kB	0.43									23.30 ± 0.13	c
WFPC2 F450W	0.45			20.26 ± 0.04	This Work						
SDSS g	0.47	21.51 ± 0.17	This Work	19.95 ± 0.09	This Work	21.08 ± 0.10	This Work	19.74 ± 0.09	This Work		
WFC3 F450W	0.47							19.86 ± 0.05	This Work		
CFHT gs	0.48					21.08 ± 0.10	a				
WFPC2 F550W	0.54					20.70 ± 0.05	This Work				
CFHT vs	0.55					20.56 ± 0.10	a				
WFC3 F606W	0.58							19.63 ± 0.05	This Work		
ACS F606W	0.58	20.44 ± 0.05	This Work	19.69 ± 0.05	This Work						
SDSS r	0.62	20.09 ± 0.09	This Work	19.46 ± 0.09	This Work	20.45 ± 0.10	This Work	19.36 ± 0.09	This Work		
CFHT rs	0.62					20.36 ± 0.10	a				
CFHT 12kR	0.66									21.99 ± 0.02	c
WFPC2 F675W	0.67					20.40 ± 0.05	This Work				
SDSS i	0.75	19.55 ± 0.10	This Work	19.15 ± 0.10	This Work	20.29 ± 0.10	This Work	18.96 ± 0.10	This Work		
CFHT is	0.77					20.31 ± 0.10	a				
WFC3 F814W	0.80							19.36 ± 0.05	This Work		
ACS F814W	0.80	19.89 ± 0.05	This Work	19.45 ± 0.05	This Work	20.38 ± 0.05	This Work			21.60 ± 0.05	This Work
SDSS z	0.89	19.29 ± 0.10	This Work	18.99 ± 0.10	This Work	20.42 ± 0.10	This Work	< 20.77	This Work		
NICMOS F110W	1.10	19.97 ± 0.05	This Work	19.44 ± 0.05	This Work						
WFC3 F110W	1.14							18.98 ± 0.06	This Work		
VLT J	1.23									20.10 ± 0.03	c
IRTF J	1.25					19.93 ± 0.10	b				
WFC3 F160W	1.53							18.56 ± 0.05	This Work		
NICMOS F160W	1.59	19.24 ± 0.05	This Work	19.07 ± 0.05	This Work	19.79 ± 0.05	This Work				
CFHT H	1.63					19.81 ± 0.10	b				
VLT H	1.65									19.77 ± 0.04	c
IRTF Kp	2.11					19.60 ± 0.10	b				
UKIRT K	2.20									19.47 ± 0.04	c
IRAC Ch1	3.54	18.12 ± 0.15	This Work	18.38 ± 0.15	This Work	19.08 ± 0.15	This Work	18.17 ± 0.15	This Work	18.58 ± 0.20	This Work
IRAC Ch2	4.48	18.20 ± 0.15	This Work	18.12 ± 0.15	This Work	19.05 ± 0.15	This Work	18.27 ± 0.15	This Work	18.49 ± 0.20	This Work
IRAC Ch3	5.70			18.32 ± 0.15	This Work	19.28 ± 0.15	This Work	18.45 ± 0.15	This Work		
IRAC Ch4	7.80	18.32 ± 0.15	This Work	18.29 ± 0.15	This Work	19.39 ± 0.15	This Work	18.91 ± 0.15	This Work		

References— [a] H. K. C. Yee et al. (1996), [b] E. Ellingson et al. (1996), [c] P. Sklias et al. (2014)

Table 11. IR photometry used for spectrum fitting

		Cosmic Eye		8:00arc		cB58		Cosmic Horseshoe		A68-h7	
Filter	Wavelength	Flux	Ref.	Flux	Ref.	Flux	Ref.	Flux	Ref.	Flux	Ref.
	μm	(mJy)		(mJy)		(mJy)		(mJy)		(mJy)	
IRS 16	15.69	0.09 ± 0.01	a								
MIPS 24	23.43	0.28 ± 0.04	b							0.34 ± 0.06	e
MIPS 70	69.86	4.10 ± 1.30	a								
PACS	70.02	5.40 ± 1.30	c	15.10 ± 3.20	c	4.30 ± 1.24	c	2.20 ± 1.00	c		
	99.60	10.90 ± 2.46	c	36.80 ± 7.40	c	9.30 ± 2.11	c	6.70 ± 1.73	c	8.37 ± 0.37	e
	158.60	17.40 ± 3.50	c	63.90 ± 12.90	c	21.60 ± 4.61	c	15.20 ± 3.30	c	24.60 ± 0.95	e
SPIRE	246.10	26.80 ± 7.00	c	91.30 ± 18.50	c	24.90 ± 5.53	c	24.80 ± 6.10	c	40.20 ± 1.33	e
	345.50	33.10 ± 8.70	c	78.50 ± 16.20	c	16.00 ± 4.60	c	16.20 ± 5.46	c	27.40 ± 1.48	e
	493.10	19.10 ± 10.00	c	47.80 ± 10.40	c	10.90 ± 4.21	c	9.30 ± 4.14	c	12.00 ± 1.29	e
ALMA Band 8	650.70	< 2.89	This Work								
ALMA Band 7	849.90	2.70 ± 0.27	This Work								
JCMT SCUBA	861.30									3.26 ± 0.70	e
ALMA Band 7	873.60							2.76 ± 0.28	This Work		
	887.70					< 1.68	This Work				
MAMBO	1210.00	1.60 ± 0.50	c	2.90 ± 0.77	c	1.10 ± 0.37	d				
ALMA Band 6	1291.00					< 0.91	This Work				
	1292.00			1.91 ± 0.19	This Work						
	1299.00							< 0.18	This Work		
ALMA Band 4	2164.00					< 0.06	This Work				
	2240.00			0.30 ± 0.05	This Work						
ALMA Band 3	2781.00					< 0.01	This Work				
	2942.00			< 0.09	This Work						
	3094.00							< 0.17	This Work		
	3233.00	< 0.08	This Work								

References— [a] B. Siana et al. (2009), [b] K. E. K. Coppin et al. (2007), [c] A. Saintonge et al. (2013), [d] A. J. Baker et al. (2001), [e] P. Sklias et al. (2014)

REFERENCES

Accurso, G., Saintonge, A., Catinella, B., et al. 2017, MNRAS, 470, 4750, doi: 10.1093/mnras/stx1556

Adelberger, K. L., & Steidel, C. C. 2000, ApJ, 544, 218, doi: 10.1086/317183

Allam, S. S., Tucker, D. L., Lin, H., et al. 2007, ApJL, 662, L51, doi: 10.1086/519520

Arimoto, N., Sofue, Y., & Tsujimoto, T. 1996, PASJ, 48, 275, doi: 10.1093/pasj/48.2.275

Arriagada-Neira, S., Herrera-Camus, R., Villanueva, V., et al. 2025, A&A, 696, A83, doi: 10.1051/0004-6361/202452652

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Baker, A. J., Lutz, D., Genzel, R., Tacconi, L. J., & Lehnert, M. D. 2001, A&A, 372, L37, doi: 10.1051/0004-6361:20010576

Baker, A. J., Tacconi, L. J., Genzel, R., Lehnert, M. D., & Lutz, D. 2004, ApJ, 604, 125, doi: 10.1086/381798

Belokurov, V., Evans, N. W., Moiseev, A., et al. 2007, ApJL, 671, L9, doi: 10.1086/524948

Bigiel, F., Leroy, A., Walter, F., et al. 2008, AJ, 136, 2846, doi: 10.1088/0004-6256/136/6/2846

Blain, A. W. 1999, MNRAS, 304, 669, doi: 10.1046/j.1365-8711.1999.02426.x

Blandford, R. D., & Narayan, R. 1992, ARA&A, 30, 311, doi: 10.1146/annurev.aa.30.090192.00152310.1146/annurev.astro.30.1.311

Boissier, S., Prantzos, N., Boselli, A., & Gavazzi, G. 2003, MNRAS, 346, 1215, doi: 10.1111/j.1365-2966.2003.07170.x

Bolatto, A. D., Wolfire, M., & Leroy, A. K. 2013, ARA&A, 51, 207, doi: 10.1146/annurev-astro-082812-140944

Bournaud, F., Perret, V., Renaud, F., et al. 2014, ApJ, 780, 57, doi: 10.1088/0004-637X/780/1/57

Cañameras, R., Nesvadba, N., Kneissl, R., et al. 2017, A&A, 604, A117, doi: 10.1051/0004-6361/201630186

CASA Team, Bean, B., Bhatnagar, S., et al. 2022, PASP, 134, 114501, doi: 10.1088/1538-3873/ac9642

Chen, Y.-C., Ishikawa, Y., Zakamska, N. L., et al. 2024, ApJ, 968, 92, doi: 10.3847/1538-4357/ad4798

Chiang, I.-D., Sandstrom, K. M., Chastenet, J., et al. 2024, ApJ, 964, 18, doi: 10.3847/1538-4357/ad23ed

Coppin, K. E. K., Swinbank, A. M., Neri, R., et al. 2007, ApJ, 665, 936, doi: 10.1086/519789

Daddi, E., Elbaz, D., Walter, F., et al. 2010, ApJL, 714, L118, doi: 10.1088/2041-8205/714/1/L118

Decarli, R., Smail, I., Walter, F., et al. 2014, ApJ, 780, 115, doi: 10.1088/0004-637X/780/2/115

Decarli, R., Walter, F., Aravena, M., et al. 2016, ApJ, 833, 70, doi: 10.3847/1538-4357/833/1/70

Dessauges-Zavadsky, M., Zamojski, M., Schaerer, D., et al. 2015, A&A, 577, A50, doi: 10.1051/0004-6361/201424661

Downes, D., & Solomon, P. M. 1998, ApJ, 507, 615, doi: 10.1086/306339

Downes, D., & Solomon, P. M. 2003, ApJ, 582, 37, doi: 10.1086/344594

Downes, D., Solomon, P. M., & Radford, S. J. E. 1993, ApJL, 414, L13, doi: 10.1086/186984

Draine, B. T., & Li, A. 2007, ApJ, 657, 810, doi: 10.1086/511055

Dye, S., Smail, I., Swinbank, A. M., Ebeling, H., & Edge, A. C. 2007, MNRAS, 379, 308, doi: 10.1111/j.1365-2966.2007.11960.x

Ellingson, E., Yee, H. K. C., Bechtold, J., & Elston, R. 1996, ApJL, 466, L71, doi: 10.1086/310182

Elmegreen, B. G. 2002, ApJ, 577, 206, doi: 10.1086/342177

Fernández Aranda, R., Díaz Santos, T., Hatziminaoglou, E., et al. 2025, A&A, 695, L15, doi: 10.1051/0004-6361/202453214

Fixsen, D. J., Bennett, C. L., & Mather, J. C. 1999, ApJ, 526, 207, doi: 10.1086/307962

Förster Schreiber, N. M., & Wuyts, S. 2020, ARA&A, 58, 661, doi: 10.1146/annurev-astro-032620-021910

Förster Schreiber, N. M., Shapley, A. E., Genzel, R., et al. 2011, ApJ, 739, 45, doi: 10.1088/0004-637X/739/1/45

Gao, Y., & Solomon, P. M. 2004, ApJ, 606, 271, doi: 10.1086/382999

Genzel, R., Tacconi, L. J., Gracia-Carpio, J., et al. 2010, MNRAS, 407, 2091, doi: 10.1111/j.1365-2966.2010.16969.x

Genzel, R., Newman, S., Jones, T., et al. 2011, ApJ, 733, 101, doi: 10.1088/0004-637X/733/2/101

Genzel, R., Tacconi, L. J., Combes, F., et al. 2012, ApJ, 746, 69, doi: 10.1088/0004-637X/746/1/69

Genzel, R., Tacconi, L. J., Kurk, J., et al. 2013, ApJ, 773, 68, doi: 10.1088/0004-637X/773/1/68

Gómez, J. S., Messias, H., Nagar, N. M., et al. 2025, A&A, 704, A178, doi: 10.1051/0004-6361/202554705

Greve, T. R., Leonidaki, I., Xilouris, E. M., et al. 2014, ApJ, 794, 142, doi: 10.1088/0004-637X/794/2/142

Hainline, K. N., Shapley, A. E., Kornei, K. A., et al. 2009, ApJ, 701, 52, doi: 10.1088/0004-637X/701/1/52

Hao, C.-N., Kennicutt, R. C., Johnson, B. D., et al. 2011, ApJ, 741, 124, doi: 10.1088/0004-637X/741/2/124

Hezaveh, Y. D., Marrone, D. P., & Holder, G. P. 2012, ApJ, 761, 20, doi: 10.1088/0004-637X/761/1/20

Hodge, J. A., Riechers, D., Decarli, R., et al. 2015, ApJL, 798, L18, doi: 10.1088/2041-8205/798/1/L18
Hodge, J. A., Swinbank, A. M., Simpson, J. M., et al. 2016, ApJ, 833, 103, doi: 10.3847/1538-4357/833/1/103
Högbom, J. A. 1974, A&AS, 15, 417
Horie, S., Okamoto, T., & Habe, A. 2024, MNRAS, 527, 10077, doi: 10.1093/mnras/stad3798
Ivison, R. J., Papadopoulos, P. P., Smail, I., et al. 2011, MNRAS, 412, 1913, doi: 10.1111/j.1365-2966.2010.18028.x
Johnson, B. D., Leja, J., Conroy, C., & Speagle, J. S. 2021, ApJS, 254, 22, doi: 10.3847/1538-4365/abef67
Kennicutt, Jr., R. C. 1998a, ApJ, 498, 541, doi: 10.1086/305588
Kennicutt, Jr., R. C. 1998b, ARA&A, 36, 189, doi: 10.1146/annurev.astro.36.1.189
Kennicutt, Jr., R. C., & De Los Reyes, M. A. C. 2021, ApJ, 908, 61, doi: 10.3847/1538-4357/abd3a2
Kennicutt, Jr., R. C., Calzetti, D., Walter, F., et al. 2007, ApJ, 671, 333, doi: 10.1086/522300
Kirkpatrick, A., Sharon, C., Keller, E., & Pope, A. 2019, ApJ, 879, 41, doi: 10.3847/1538-4357/ab223a
Kroupa, P. 2001, MNRAS, 322, 231, doi: 10.1046/j.1365-8711.2001.04022.x
Krumholz, M. R., & McKee, C. F. 2005, ApJ, 630, 250, doi: 10.1086/431734
Krumholz, M. R., & Thompson, T. A. 2007, ApJ, 669, 289, doi: 10.1086/521642
Larson, R. B. 1981, MNRAS, 194, 809, doi: 10.1093/mnras/194.4.809
Leja, J., Johnson, B. D., Conroy, C., van Dokkum, P. G., & Byler, N. 2017, ApJ, 837, 170, doi: 10.3847/1538-4357/aa5ffe
Leroy, A., Bolatto, A., Walter, F., & Blitz, L. 2006, ApJ, 643, 825, doi: 10.1086/503024
Leroy, A., Bolatto, A. D., Simon, J. D., & Blitz, L. 2005, ApJ, 625, 763, doi: 10.1086/429578
Leroy, A. K., Walter, F., Brinks, E., et al. 2008, AJ, 136, 2782, doi: 10.1088/0004-6256/136/6/2782
Leroy, A. K., Bolatto, A., Gordon, K., et al. 2011, ApJ, 737, 12, doi: 10.1088/0004-637X/737/1/12
Leroy, A. K., Walter, F., Sandstrom, K., et al. 2013, AJ, 146, 19, doi: 10.1088/0004-6256/146/2/19
Li, J., Da Cunha, E., González-López, J., et al. 2024, ApJ, 976, 70, doi: 10.3847/1538-4357/ad7fee
Li, Z., Cai, Z., Wang, X., et al. 2025, ApJS, 280, 62, doi: 10.3847/1538-4365/adfa70
Lilly, S. J., Le Fevre, O., Hammer, F., & Crampton, D. 1996, ApJL, 460, L1, doi: 10.1086/309975
Lindner, R. R., Baker, A. J., Beelen, A., Owen, F. N., & Polletta, M. 2012, ApJ, 757, 3, doi: 10.1088/0004-637X/757/1/3
Madau, P., Pozzetti, L., & Dickinson, M. 1998, ApJ, 498, 106, doi: 10.1086/305523
Madau, P., & Shull, J. M. 1996, ApJ, 457, 551, doi: 10.1086/176751
Magnelli, B., Saintonge, A., Lutz, D., et al. 2012, A&A, 548, A22, doi: 10.1051/0004-6361/201220074
McMaster, M., & et al. 2008, Wide Field and Planetary Camera 2 Instrument Handbook v. 10.0
Meier, D. S., Turner, J. L., Beck, S. C., et al. 2010, The Astronomical Journal, 140, 1294
Murgia, M., Crapsi, A., Moscadelli, L., & Gregorini, L. 2002, A&A, 385, 412, doi: 10.1051/0004-6361:20020140
Narayanan, D., Cox, T. J., Shirley, Y., et al. 2008, ApJ, 684, 996, doi: 10.1086/588720
Narayanan, D., Krumholz, M., Ostriker, E. C., & Hernquist, L. 2011, MNRAS, 418, 664, doi: 10.1111/j.1365-2966.2011.19516.x
Ostriker, E. C., & Shetty, R. 2011, ApJ, 731, 41, doi: 10.1088/0004-637X/731/1/41
Rawle, T. D., Egami, E., Bussmann, R. S., et al. 2014, ApJ, 783, 59, doi: 10.1088/0004-637X/783/1/59
Reddy, N. A., Erb, D. K., Pettini, M., Steidel, C. C., & Shapley, A. E. 2010, ApJ, 712, 1070, doi: 10.1088/0004-637X/712/2/1070
Richard, J., Jones, T., Ellis, R., et al. 2011, MNRAS, 413, 643, doi: 10.1111/j.1365-2966.2010.18161.x
Riechers, D. A., Carilli, C. L., Walter, F., & Momjian, E. 2010, ApJL, 724, L153, doi: 10.1088/2041-8205/724/2/L153
Saintonge, A., Lutz, D., Genzel, R., et al. 2013, ApJ, 778, 2, doi: 10.1088/0004-637X/778/1/2
Sandstrom, K. M., Leroy, A., Walter, F., et al. 2013, The Astrophysical Journal, 777, 5
Sargent, M. T., Daddi, E., Béthermin, M., et al. 2014, ApJ, 793, 19, doi: 10.1088/0004-637X/793/1/19
Schmidt, M. 1959, ApJ, 129, 243, doi: 10.1086/146614
Schuster, K. F., Kramer, C., Hitschfeld, M., Garcia-Burillo, S., & Mookerjea, B. 2007, A&A, 461, 143, doi: 10.1051/0004-6361:20065579
Scoville, N. Z., Sargent, A. I., Sanders, D. B., & Soifer, B. T. 1991, ApJL, 366, L5, doi: 10.1086/185897
Seitz, S., Saglia, R. P., Bender, R., et al. 1998, MNRAS, 298, 945, doi: 10.1046/j.1365-8711.1998.01443.x
Semenov, V. A., Kravtsov, A. V., & Gnedin, N. Y. 2019, ApJ, 870, 79, doi: 10.3847/1538-4357/aaf163
Serjeant, S. 2012, MNRAS, 424, 2429, doi: 10.1111/j.1365-2966.2012.20761.x

Sharda, P., Wisnioski, E., Krumholz, M. R., & Federrath, C. 2021, MNRAS, 506, 1295, doi: 10.1093/mnras/stab1836
Sharon, C. E., Baker, A. J., Harris, A. I., & Thomson, A. P. 2013, ApJ, 765, 6, doi: 10.1088/0004-637X/765/1/6
Sharon, C. E., Tagore, A. S., Baker, A. J., et al. 2019, ApJ, 879, 52, doi: 10.3847/1538-4357/ab22b9
Siana, B., Smail, I., Swinbank, A. M., et al. 2009, ApJ, 698, 1273, doi: 10.1088/0004-637X/698/2/1273
Sklias, P., Zamojski, M., Schaerer, D., et al. 2014, A&A, 561, A149, doi: 10.1051/0004-6361/201322424
Smail, I., Swinbank, A. M., Richard, J., et al. 2007, ApJL, 654, L33, doi: 10.1086/510902
Solomon, P. M., Downes, D., & Radford, S. J. E. 1992, ApJL, 398, L29, doi: 10.1086/186569
Solomon, P. M., Downes, D., Radford, S. J. E., & Barrett, J. W. 1997, ApJ, 478, 144, doi: 10.1086/303765
Solomon, P. M., Rivolo, A. R., Barrett, J., & Yahil, A. 1987, ApJ, 319, 730, doi: 10.1086/165493
Solomon, P. M., & Sage, L. J. 1988, ApJ, 334, 613, doi: 10.1086/166865
Somerville, R. S., & Davé, R. 2015, ARA&A, 53, 51, doi: 10.1146/annurev-astro-082812-140951
Strong, A. W., & Mattox, J. R. 1996, A&A, 308, L21
Suchkov, A. 2001, in ACS Instrument Handbook for Cycle 11 v. 2.0, Vol. 2, 2
Tadaki, K., Iono, D., Yun, M. S., et al. 2018, Nature, 560, 613, doi: 10.1038/s41586-018-0443-1
Teng, Y.-H., Sandstrom, K. M., Sun, J., et al. 2023, ApJ, 950, 119, doi: 10.3847/1538-4357/accb86
Teplitz, H. I., McLean, I. S., Becklin, E. E., et al. 2000, ApJL, 533, L65, doi: 10.1086/312595
Tsukui, T., Wisnioski, E., Krumholz, M. R., & Battisti, A. 2023, MNRAS, 523, 4654, doi: 10.1093/mnras/stad1464
Vallini, L., Witstok, J., Sommovigo, L., et al. 2024, MNRAS, 527, 10, doi: 10.1093/mnras/stad3150
Vizgan, D., Vieira, J. D., Spilker, J. S., et al. 2026, ApJ, 999, 118, doi: 10.3847/1538-4357/ae3744
Walter, F., Decarli, R., Sargent, M., et al. 2014, ApJ, 782, 79, doi: 10.1088/0004-637X/782/2/79
Wang, B., Leja, J., Bezanson, R., et al. 2023, ApJL, 944, L58, doi: 10.3847/2041-8213/acba99
Wang, B., Leja, J., Atek, H., et al. 2024, ApJ, 963, 74, doi: 10.3847/1538-4357/ad187c
Wei, L. H., Vogel, S. N., Kannappan, S. J., et al. 2010, ApJL, 725, L62, doi: 10.1088/2041-8205/725/1/L62
Whitaker, K. E., van Dokkum, P. G., Brammer, G., & Franx, M. 2012, ApJL, 754, L29, doi: 10.1088/2041-8205/754/2/L29
Wilson, C. D. 1995, ApJL, 448, L97, doi: 10.1086/309615
Wong, M. H., Pavlovsky, C., & Long, K. S. 2010, in WFC3 Instrument Handbook for Cycle 18 v. 2.0, Vol. 2, 2
Wong, T., & Blitz, L. 2002, ApJ, 569, 157, doi: 10.1086/339287
Wuyts, S., Förster Schreiber, N. M., van der Wel, A., et al. 2011, ApJ, 742, 96, doi: 10.1088/0004-637X/742/2/96
Yee, H. K. C., Ellingson, E., Bechtold, J., Carlberg, R. G., & Cuillandre, J.-C. 1996, AJ, 111, 1783, doi: 10.1086/117916
Yuan, T.-T., Kewley, L. J., & Richard, J. 2013, ApJ, 763, 9, doi: 10.1088/0004-637X/763/1/9
Zamora-Avilés, M., Vázquez-Semadeni, E., & Colín, P. 2012, ApJ, 751, 77, doi: 10.1088/0004-637X/751/1/77
Zanella, A., Iani, E., Dessauges-Zavadsky, M., et al. 2024, A&A, 685, A80, doi: 10.1051/0004-6361/202349074
Zhu, Y., Rieke, M. J., Ji, Z., et al. 2026, ApJ, 1000, 303, doi: 10.3847/1538-4357/ae50f1